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Asymmetric photocatalytic reduction by in-situ transfer hydrogenation using metallosalan-MOFs

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Civilingenjörsprogrammet i kemiteknik



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Abstract

A multi-step synthetic column-free route was developed for a cobalt(III) salan complex (CoSalan) with carboxylic acid moeities, designed to serve as a bi-functional chiral linker in metal-organic framework (MOF) synthesis. The ultimate goal was to incorporate this linker into a MOF and evaluate its performance as a heterogeneous photo-catalyst for asymmetric transfer hydrogenation (ATH) measuring its enantioselectivity on different substrates.

The synthesis proceeded through seven steps from 2-*tert*-butylphenol: Friedel-Crafts acylation, pyridinium iodide salt formation and subsequent hydrolysis yielding a carboxylic acid group, Duff-type ortho-formylation, Schiff base condensation with *trans*-1,2-cyclohexanediamine to give salen-type ligand, $NaBH_4$ reduction to the hydrolytically stable salan in the form of a HCl salt, and cobalt complexation to yield CoSalan complex. All intermediates were characterized by ^{1}H NMR and some of them with ^{13}C NMR.

The CoSalan linker was successfully synthesized; FTIR data indicate cobalt complexation, evidenced by a shift of the N-H stretching vibration from 3220 cm^{-1} to 3410 cm^{-1} compared to the salan HCl salt. Characterization by cyclic voltammetry, EPR, and SCXRD for CoSalan was planned but not completed within the project timeline. MOF synthesis was not achieved. The principal bottlenecks were persistent iodide contamination from an earlier step, sensitivity of the Schiff base condensation to temperature and scale, and demanding purification of the salan HCl salt. Practical recommendations for each bottleneck are provided as a basis for continuation of this work.

0.1 Preface

AI was used to revise and improve the existing text linguistically.

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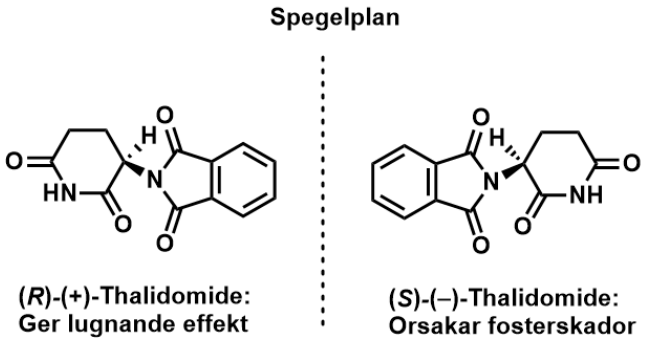
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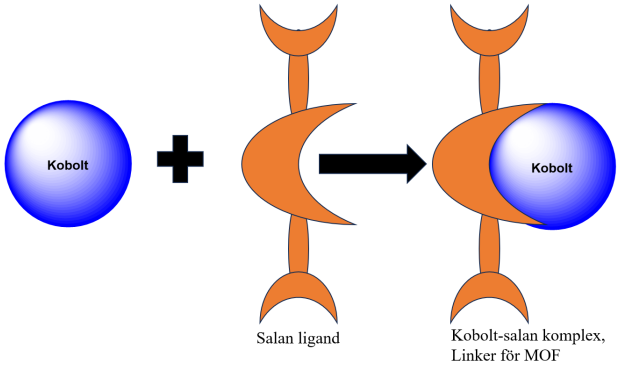
Populärvetenskaplig sammanfattning

I det här arbetet tillverkades en funktionell ”linker” i sju steg utan att behöva renas med kolonn-kromatografi, vilket är attraktivt ur ett hållbarhetsaspekt eftersom kolonnkromatografi kräver enorma mängder lösningsmedel som blir svinn. Tanken är att den senare ska kunna användas för att bilda en ”MOF”. MOF står för **Metal organic framework** och består utav metallbaserade knutpunkter som sitter sammanlänkade med dessa linkers. Beroende på vilka byggstenar som väljs kan MOFar designas för en mängd olika tillämpningar, däribland katalys, gasseparation och läkemedelsleverans. 2025 vann MOFar Nobelpriset tack vare deras unika egenskaper hos funktionella material var det gäller porösitet med stora hålrum, stabilitet som utvecklats och förmåga att kunna skräddarsys för olika uppgifter med rationell design.

I detta fallet så ska MOFen kunna katalysera specifika kemiska reaktioner där man bara vill ha en utav 2 möjliga spegelvända molekyler som produkt. Den specifika kemiska reaktionen som skall nyttjas kallas för **Asymmetric transfer hydrogenation** (ATH), där ”asymmetric” betyder att den är selektiv för ena spegelbildsvarianten. Sådana asymmetriskt selektiva reaktioner är centrala inom läkemedelsindustrin, eftersom olika spegelbildsvarianter av samma läkemedelsmolekyl kan ha dramatiskt olika biologisk aktivitet – ett känt exempel är talidomidkatastrofen under 1950- och 60-talen, där talomid gavs till bl.a. gravida patienter för morgonsjuka. Vad man inte visste var att *(S)*-varienten orsakade fosterskador, men det framstod sen vid födseln då barnen hade ofärdigt bildade armar och ben.



att MOFens katalytiska funktion skulle kunna bli aktiverad av att synligt ljus skiner på den vilket ger den energin för att utföra sitt arbete (Dess arbete är själva katalyseringen av den kemiska reaktionen) istället för värme. Detta skulle i det här fallet ske på självaste linkern som skapades i detta arbetet. Dock så måste även ”ground state reactivity” undersökas, vilkät är om ljus inte skulle behövas alls. Linkern är ett metall-komplex bestående av kobolt som omringas av en organisk molekyl (salan) som håller fast den. Denna typ av komplex kan för sig själv katalysera den specifika kemiska reaktionen som nämndes i tidigare stycket, ATH. ”Transfer hydrogenation” betyder att metallcentret (i detta fallet kobolt) antar en negativt laddad väteatom som kallas för hydrid. Denna hydrid överförs sedan till substratet som ska ha denna reaktion utövad på sig, vilket kan skapa två spegelbildsvarienter. På grund av hur linker-komplexet är designat så kommer den ena önskade varienten att skapas med fördel. Till vilken grad detta sker är något som måste utvärderas.



Poängen med att sätta fast kobolt-salan-komplexet i en MOF som en linker är att öka effektiviteten för den att utöva ATH, förbättra stabilitet och hållbarhet, så den kan återvinnas. Man sparar även energin som annars skulle krävas för dessa specifika reaktioner. Eftersom dessa reaktioner är specifika kan det göras på en känslig molekyl som startmaterial som annars hade brännts upp i fall man hetta upp den eller reagerat annorlunda och ej givit önskad produkt på grund av selektivitetsproblem. Projektet resulterade i en framgångsrik syntes av kobolt-salan-linkern, bekräftad via *Fourier Transform Infrared Spectroscopy* (FTIR), där man kan se hur en ”topp” i spektrumet skiftats mellan salen-liganden jämfört med kobolt-salan-komplexet. Detta indikerar att koboltkomplexeringen har lyckats. MOF-syntesen hann dock inte genomföras inom projektets tidsram. I det här arbetet identifieras konkreta flaskhalsar och tydliga rekommendationer för fortsätt forskning ges.

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1 Abbreviations

MOF	Metal-organic framework
PSM	Post synthetic modification
ATH	Assymetric transfer hydrogenation
HOMO	Highest occupied molecular orbital
LUMO	Lowest unoccupied molecular orbital
LCMT	Ligand-metal charge transfer
SET	Single electron transfer
MLC	Metal-ligand cooperation
MIL	Materials of Institute Lavoisier
SBU	Secondary building unit
EtOH	Ethanol
MeOH	Methanol
Et_2O	Diethyl ether
TFA	Trifluoroacetic acid
MeCN	Acetonitrile
DACH	*trans*-1,2-cyclohexanediamine
EtOAc	Ethyl acetate
$Co(OAc)_2$	Cobalt acetate
FTIR	Fourier transformed infrared spectroscopy
NMR	Nuclear magnetic resonance
EPR	Electroparamagnetic resonance spectroscopy
SCXRD	Single crystal X-ray diffraction
MS	Mass spectrometry
TGA	Thermogravimetric analysis
PXRD	Powder X-ray diffraction
SEM	Scanning electron microscopy

2 Introduction

Developing the field of enantioselective heterogeneous catalysts is of great interest interdisciplinary[1] since it could increase the efficiency of organic reactions, such as oxidations, reductions, epoxidations from the point of view of sustainability[2]. In this project, the end goal was to examine catalysis of the asymmetric transfer hydrogenation (ATH) reaction, illustrated in figure 1. This reaction is one of the most important for organic synthesis, especially when it comes to synthesizing drugs with chiral centers since mirror-image molecules can have entirely different pharmacological profiles. The thalidomide tragedy of the 1950s and 1960s, in which one enantiomer caused severe teratogenic effects while the other was therapeutically active, exemplifies the safety imperative for enantiomerically pure compounds[3]. These catalysts could also be used for bio-mass derived feedstocks and degradation of wastewater pollutants through a Fenton like mechanism photocatalytically[4;5]. On complex substrates in the fine chemical industries, developing a milder reaction conditions is heavily sought after as it would allow for new synthetic manipulations or up-scaling in industry.

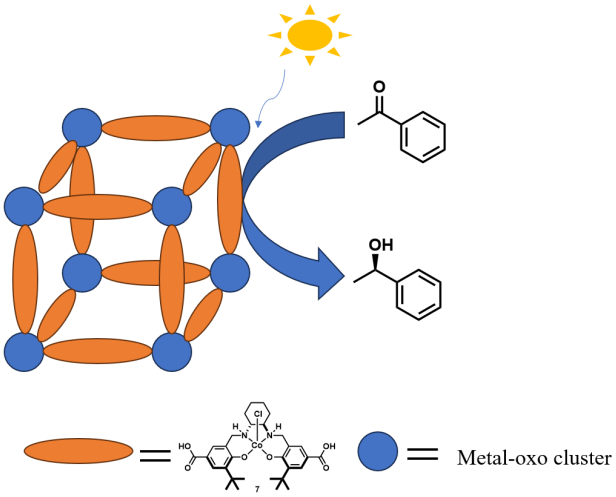


Figure 1: Illustration of the project aim: a MOF containing the CoSalan linker (7) catalyses ATH of a pro-chiral ketone to a chiral alcohol using visible light as the energy source.

Metal-salen complexes have been used both as homogeneous[6;7;8] and heterogeneous catalysts[9;10] before with different metals. They show great potential at being immobilized in MOFs, which yields heterogeneous catalysts that combine the enantioselectivity of molecular catalysts with the practical advantages of solid-phase materials: ease of separation, recyclability, and high surface area[9;10;11;12]. MOFs were recognised with the 2025 Nobel Prize in Chemistry and have been developed since 1960s with modern MOFs as we know them today first being developed in the 1990s[13]. Among heterogeneous catalysts, MOF catalysts show comparatively great solvent and thermal stability[14;15]. They also are very easily modifiable through post synthetic modification (PSM)[16] such that one can tune their band gap, photo chemistry, conductivity and more.[17;18]. A particularly attractive approach is photocatalytic ATH, in which the catalyst is activated by visible light rather than heat, avoiding side reactions with sensitive substrates and eliminating the need for external flammable H_2 gas, using a sacrificial donor (an alcohol) as the hydrogen source improving selectivity, atom economy, safety and sustainability[19]. For ATH reactions at a small scale, Chiral Ru(II), Ir(II) and Mn complexes ar widely used[20;21] which are homogeneous catalysts facing challenges of sustainability and durability at larger scales. There have been many attempts to use transition metals and also with making them heterogeneous to perform this reaction and improve atom economy[9;2;10]. Developing a more sustainable or novel catalytic approach is of interest and some developments[22] have found that one can substitute Ruthenium, a rare expensive platina group transition metal, with cobalt, a transition metal that is more abundant in earth's crust. When it comes to choice of metal in the metal-salen complex, in aluminum and transition metal complexes are well established in literature for their photo chemistry and LCMT band gap excitation[6] [23]. In this project, a metal-salan complex **(7)**, the reduced form of salen, salan is examined. They remain underexplored as heterogeneous catalyst except for a 2D-network[24] but have yet to be incorporated into any MOFs, likely due to their disagreeable physio-chemical properties.

2.1 Project aims

The aim of this project was to synthesise a rationally designed CoSalan complex **(7)** for use as a chiral bifunctional linker in direct MOF synthesis without the use of column chromatography for purification. The choice of Metal to the MOF is one that is structurally sound and that makes sufficient pore sizes and geometry. Therefore Zn, Al will be used. Bi will be used since they are very underexplored. and subsequently to evaluate the resulting MOF as a heterogeneous photocatalyst

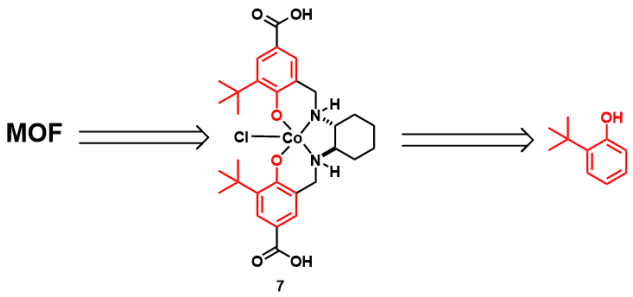


Figure 2: Retrosynthetic analysis of MOF with CoSalan complex **(7)** being synthesized from 2-*tert*-butylphenol

for ATH. The overall retrosynthetic strategy is shown in figure 2: starting from 2-*tert*-butylphenol, a seven step synthesis leads to CoSalan complex **(**7), which is incorporated into the MOF via its carboxylic acid groups coordinating to metal-oxo cluster nodes. Each intermediate compound in the seven step synthesis are illustrated in figure 3. Compound **(1)** is the Friedel-Crafts product that serves as the starting point for carboxylic acid installation. Compounds **(2)** and **(3)** are the pyridinium iodide salt intermediate and the resulting carboxylic acid for MOF node coordination. Compound **(4)** introduces the aldehyde required for Schiff base condensation by Duff-type orthoformylation. Compound **(5)** is the bis-imine salen ligand formed by condensation of **(4)** with *trans*-DACH, introducing the chiral cyclohexyl backbone. Compound **(6)** is the salan dihydrochloride obtained by reduction of the imine bonds to amines, improving its hydrolytic stability. Compound **(7)** is the target CoSalan complex.

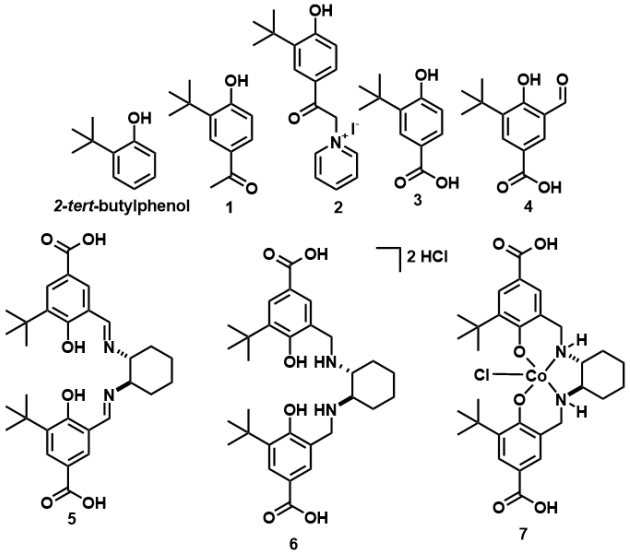


Figure 3: Structures of CoSalan complex **(7)** and all synthetic precursors **(1-6)**, starting from 2-tert-butylphenol.

The CoSalan complex **(7)** was rationally designed by adding *tert*-butyl substituents for improved solubility and steric bulk around the metal centre and inserting a *trans*-1,2-cyclohexanediamine backbone (Figure 4) introducing a defined chiral environment required for the ATH reaction to bias toward one product enantiomer. It also has carboxylic acid groups attached para to the phenolic hydroxyls as MOF coordination handles. The salan (amine) form was chosen over salen (imine) to give hydrolytic stability under the protic ATH conditions required for alcohol-donor hydrogen transfer. Cobalt(III), with its d^6-shell was chosen as the metal for its electronic properties and spin configuration, to avoid the alkoxide trap that occurs when the metal-alkoxide intermediate is stabilized from empty d-orbitals in the metal interacting with the oxygen's lone pair[25].

CoSalan Complex **(7)** can itself catalyse ATH reactions as a homogeneous catalyst[7], but MOF incorporation was intended to increase stability, facilitate recycling, and create the enforced chiral environment needed for high enantioselectivity. The subsequent planned steps were: solvothermal MOF synthesis with various metal-oxo SBUs, different conditions and ratios; structural characterisation by PXRD, TGA, and gas sorption analysis; and mechanistic investigation using TEMPO and UV-vis spectroscopy, such as if it follows a redox cycle or can perform catalysis at ground state without shining light on it. then ATH scope and enantioselectivity evaluation with pro-chiral ketones and sulfonamides was planned. As described below, the linker synthesis was completed but MOF synthesis was not achieved within the project timeline and characterization with SCXRD, UV-Vis spectroscopy, XPS and EPR of **(7)** should have been performed. A method

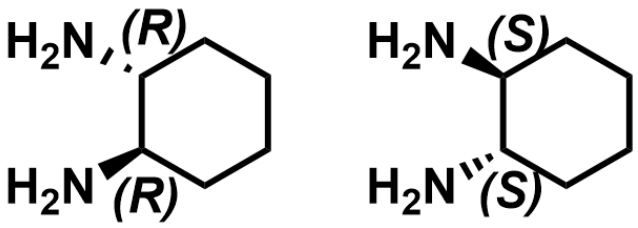


Figure 4: The structures of the two enantiomers of *trans*-DACH

for enantiomeric separation of *trans*-DACH should be developed to reach a chiral MOF but the racemic MOF will be synthesized first to save resources.

3 Theory

3.1 Photochemistry

Chemical reactions are normally driven by heating which supplies the kinetic energy needed to reach the barrier of activation and catalysts lower this barrier. Using light to catalyze a chemical reaction could be very energy efficient since the energy would be supplied in a highly targeted way instead of as bulk thermal energy with a lot of heat loss. This means that more thermally labile substrates may be used as well[26]. In general for photo chemical reactions, one reactant absorbs a photon of light which excites it from a ground singlet state S_0 to an excited singlet state S_1, from which it then may undergo an inter system crossing (ISC) to an excited triplet state T_1 (Figure 5), corresponding to HOMO-to-LUMO promotion. This can happen in the Metal-Salan catalyst as salen-type ligands are privileged, stabilizing high spin systems with ground-state reactivity augmented by excited state mechanisms[27]. Cobalt could reach a triplet state at ground state which would introduce new types of reactions, and it has proven its utility in hydrogen borrowing reactions[28]. Triplet state metal organic complexes have been examined and shown to be essential when it comes to reactivity and longevity of triplet state species[29]. When it comes to cobalt whether it has high or low spin will determine its reactivity, specificity and mechanism[30].

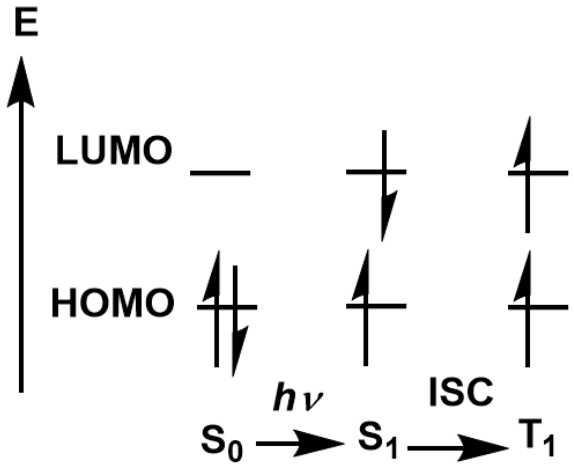


Figure 5: Orbital diagram showing excitation by photon and inter system crossing to triplet state

In metal-salen/salan complexes, coordination to the metal lowers the energy gap between occupied and unoccupied orbitals, enabling ligand-metal charge transfer (LCMT) excitation at visible wavelengths because the ligand is redshifted. Metal-salen complexes exhibit intense emission because of the LCMT band excitation and have previously been examined[6], although as homogeneous catalysts. The LCMT excitation proceeds as a π/π* transition followed by a single electron transfer (SET) which reduces the metal and the ligand becomes a radical that then quenched by a sacrificial donor, generating the metal hydride needed for ATH (Figure 6). Whether this proceeds via Fenton-type hydroxyl radicals or via direct ligand radicals remains debated but the Fenton pathway is currently considered more likely[31]. The use of visible light is important because most organic target molecules are transparent to visible radiation, whereas UV radiation can promote competitive side reactions.

3.2 Transfer hydrogenation

In ATH, there are different possible mechanisms. The first one discussed is direct hydrogenation (Figure 7 a sacrificial alcohol is oxidized to the corresponding ketone and the hydrogen is transferred to the prochiral substrate, reducing it via the activated metal complex that is then regenerated. In the cobalt salan system, the sacrificial donor acts directly as the hydride source following photo-induced reduction of Co(III) to Co(II), quenching the radical. This generates the metal hydride which is the active in the asymmetric transfer hydrogenation.

The choice of metal oxidation state and its electron configuration is critical[6;23]: for metals that has an electron configuration of d^5 or less, such as Fe(III), there will exist empty or half filled d-orbitals. This will promote d-p π "backbonding" between alkoxide intermediate formed from the sacrificial donor and the metal, known as the "alkoxide trap"[25] (Figure 8). The lone pair of the alkoxide oxygen becomes becomes trapped in the empty or half-empty d orbital of the metal, acting as a thermodynamic sink, creating an insurmountable energy barrier ultimately preventing hydride transfer and halting the catalytic cycle. Co(III) circumvents this trap as it has a d^6 configuration.

It is important to examine the different possible mechanistic pathways of hydrogenation reactions. For the adsorption of

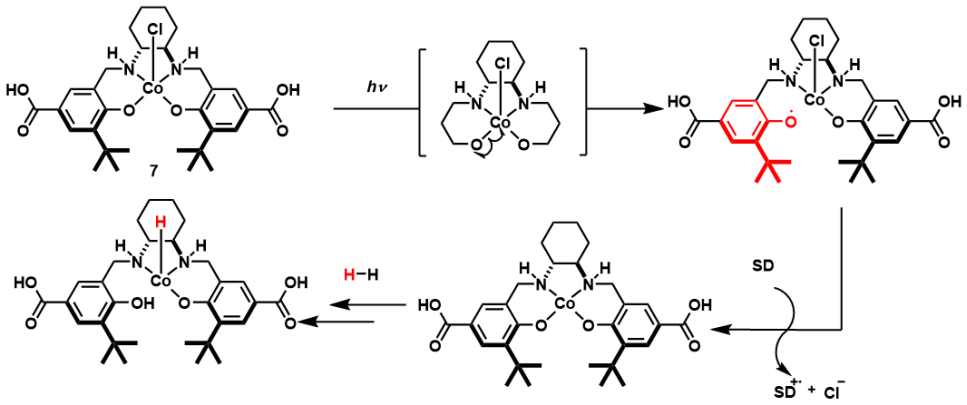


Figure 6: Photo-induced reduction of Cobalt(III) followed by hydrogenation using sacrificial donor to generate Co(II) hydride species

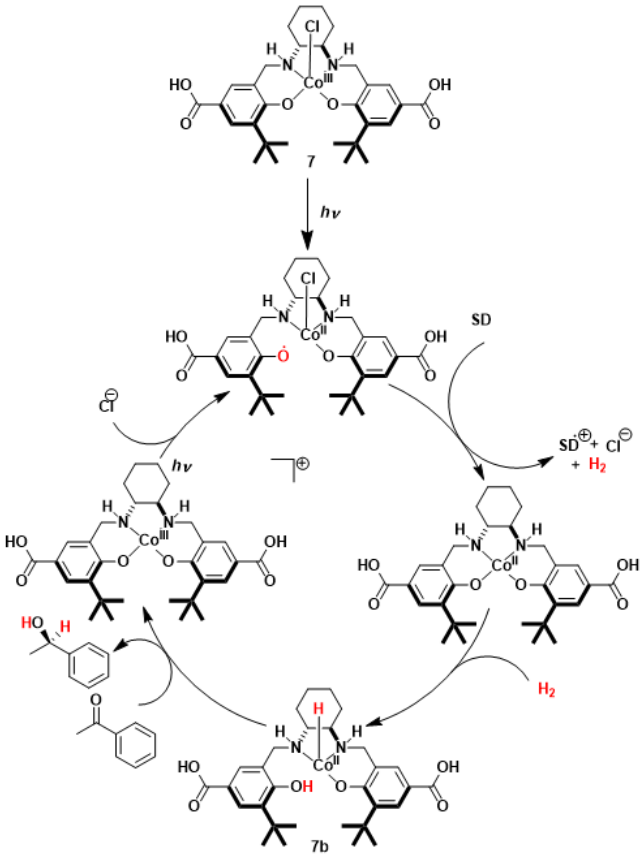


Figure 7: Direct hydrogenation catalytic cycle for **(7)** with LCMT mediated *in situ* reduction of Co(III) to Co(II) and generation of H_2 species. (speculative)

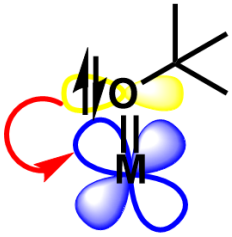


Figure 8: Illustration of the ”alkoxide trap” where the lone pair of the oxygen residing in a p-orbital interacts favorably with an empty d-orbital of a low valent Metal M in a metal complex

molecular hydrogen in generation of the activated metal complexes, there exists in general two mechanisms. Metal-ligand cooperation (MLC) is when acidic protons from the ligand are deprotonated with a base, which polarizes the H-H bond and assists in hydride formation. Ligand migratory insertion is the other one. These mechanisms do not apply for the metal-salan complexes as they may rely on a sacrificial donor that will act as the hydride. Because Co(III) was chosen, it would allow for the hydride abstraction mechanism of the sacrificial donor in the catalytic cycle (Figure 9).

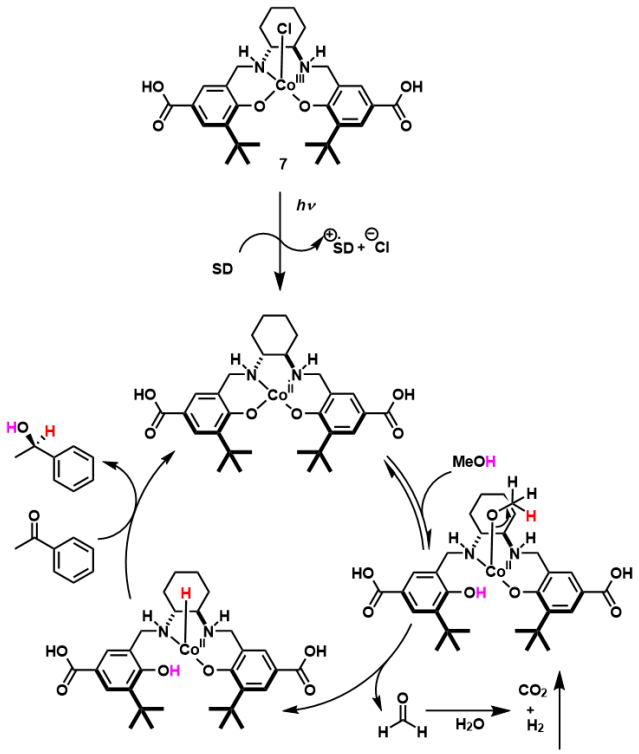


Figure 9: Catalytic cycle for **(7)** with MeOH as a sacrificial donor, highlighing alkoxide bond without alkoxide trap. (speculative)

3.3 Metal-Salan complexes

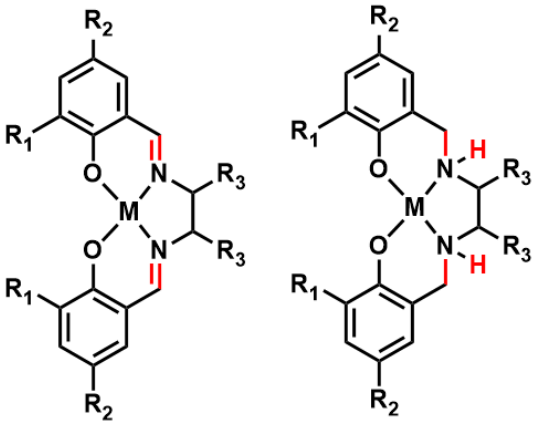


Figure 10: General Structure of M-Salen complexes (shown to the left) and M-Salan complexes (shown to the right), highlighting their differences

Metal-salen complexes consist of a cationic metal in square-planar N_2O_2 coordination. They can be synthesized using many different metals yielding compounds of differing colors, spin configuration and spin-type[32]. They can be employed as homogeneous catalysts and derivatized to increase solubility or introduce chirality. Owing to salens status as a privileged ligand, they can stabilize high spin configurations at ground state[27], which leads to their interesting reactivity[28].

The general structures of metal-salen and metal-salan are shown in figure 10, highlighting the imine versus amine coordination. Salen ligands are susceptible to imine hydrolysis in protic media[33], which is problematic for ATH with alcohol donors and for catalyst recycling. Reduction to the amine form (salan) gives hydrolytic stability at the cost of reduced π-delocalization and, in homogeneous systems, somewhat lower catalytic activity. Incorporation into a MOF is expected to recover activity through the enforced pore geometry[24]. Salan complexes are under-explored relative to salen types, making their integration into MOFs a particularly interesting research direction.

3.4 Metal-Organic Frameworks

MOFs are crystalline porous materials built from metal-containing nodes connected by organic linkers, with large surface areas and tunable pore geometries with a lot of possible applications[13;18]. For photocatalytic applications, the band gap can be tuned through PSM to match visible-light excitation[17]; MIL-type MOFs have been specifically investigated for photocatalytic activity[34]. Previous chiral salen-MOF catalysts relied on PSM strategies or partial linker exchange[9;16]; direct synthesis with a chiral metallosalan as the sole linker, as attempted here, remains largely unexplored. An important consideration for photocatalytic MOFs is light penetration: because photons cannot reach the innermost crystallites of large particles, reducing crystal size to maximise the surface-to-volume ratio is desirable for photocatalytic efficiency. Therefore to optimize the efficiency, one might want to have a higher surface area, and by extension smaller ”grain-size” of each crystal. The MOF can in theory bend and regenerate and the size of the substrate that will be allowed will be dependent upon pore-size and geometry of the MOF. The same linker could make different geometries and pore-sizes depending on the SBU’s structure and choice of metal in the metal-oxo clusters. The same metal could even yield different structures. Compared to the homogeneous metal-salen catalysts, heterogenization in a MOF which will create a better defined chiral environment that will work as a filter for racemate resolution[35].

4 Results and Discussion

The overall synthetic goal was to prepare the CoSalan complex **(7)** for direct incorporation into a MOF without the use of column chromatography for purification. The route was adapted from literature procedures[10;36;24], with a different starting point using an adapted pyridinium iodide salt synthesis **(2)** and hydrolysis[37;38] to introduce the carboxylic acid functionality for MOF node coordination. All intermediates **(1–6)** were confirmed by ^{1}H NMR (and ^{13}C NMR for **(1)** and **(3)**), with data matching literature values where available[10;39]. Salan HCl **(6)** was additionally characterised by gCOSY, gHSQC, gHMBC and FTIR (Appendix A.1, A.2). CoSalan complex **(7)** was also characterised by FTIR (Appendix A.2) to be compared with ligand **(6)**'s shifts before complexation. An alternative, more direct route via one-pot Mannich-type double alkylation of compound 3 with paraformaldehyde and *trans*-DACH was also investigated (Section 4.4). This approach would have accessed the salan ligand **(6)** in a single step but proved unsuccessful. The four-step route via salen intermediate **(5)** ultimately succeeded, though with significant optimization challenges at the condensation and reduction steps.

4.1 Synthesis of 3-tert-butyl-4-hydroxybenzoic acid, (1)

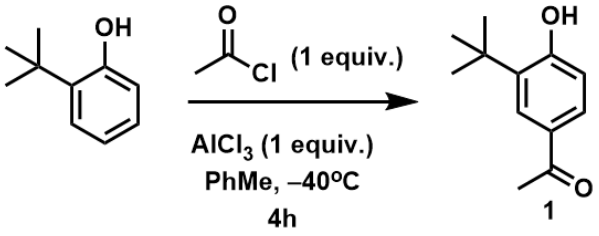


Figure 11: Synthetic scheme to obtain **(1)**: Friedel-Crafts acylation of 2-tert-butylphenol with acetyl chloride

Friedel-Crafts acylation of 2-*tert*-butyl-phenol with AcCl (1.0 equiv.) in toluene at −40°C using $AlCl_3$ gave compound 1 in 38 % (20.0 mmol scale) and 42 % (50.0 mmol scale) (Figure 11). Both were quenched in 120 ml ice water. The $AlCl_3$ that was used had a slight yellow greenish color which indicates iron(III) chloride, a common contaminant present from the process of generating $AlCl_3$ in the mining industry[40]. Moderate yields are attributed to possible partial hydration of the hygroscopic catalyst $AlCl_3$ to the hexahydrate, which would underestimate the effective molar quantity. The excessive amount of ice water used during aqueous quenching may also have contributed to product loss.

4.2 Synthesis of 1-(3-(tert-butyl)-4-hydroxyphenacyl)pyridinium iodide, (2)

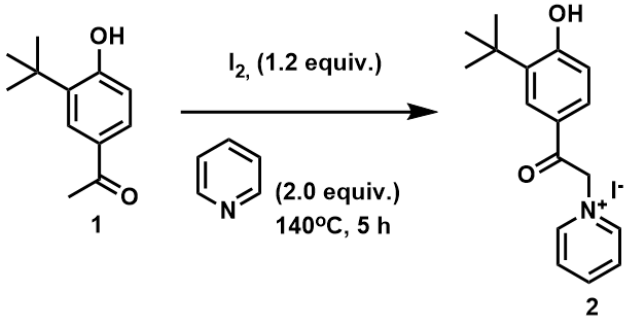


Figure 12: Synthetic scheme to obtain pyridinium iodide salt **(2)** from **(1)**, I_2 and pyridine at 140°C

Direct oxidative methods to convert the methyl ketone **(1)** to a carboxylic acid **(3)** proved unsuccessful. A two-step route via pyridinium iodide salt intermediate (2) was employed instead, adapting procedures by King[37] and Pearson et al.[41]. Treatment of **(1)** with I_2 (1.2 equiv.) in neat pyridine (2.0 equiv.) at 140°C for 5 h before recrystallization in EtOH gave compound **(2)** (Figure 12) in 84 % yield at both scales tested (7.28 mmol and 21.2 mmol of **(1)** respectively). The mechanism, analogous to the iodoform reaction with pyridine acting as base and nucleophile, is presented in Appendix B.1.

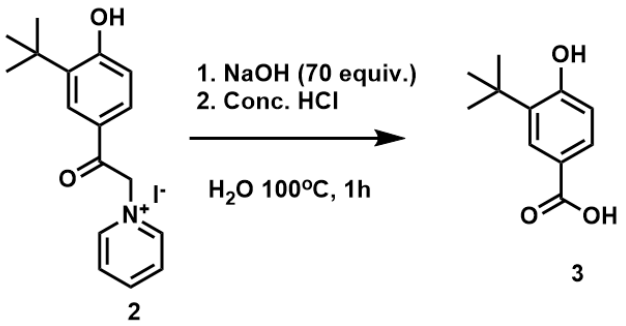


Figure 13: Synthetic scheme to obtain **(3)**: Basic hydrolysis of **(2)**.

4.3 Synthesis of 3-tert-butyl-4-hydroxybenzoic acid, (3)

Basic hydrolysis of **(2)** (NaOH, 67–90 equiv., H_2O, 100°C, 1 h) (Figure 13) followed by activated-carbon treatment to remove residual iodine species invisible in ^{1}H NMR. Then, acidification gave compound **(3)**. Optimization required six experiments (Table 1). Entry 1 failed with only 6 equiv. NaOH; the mechanistic requirement for a large excess is explained in Appendix B.2. Entries 4–6 gave 81–96 % yield using 67–70 equiv. NaOH. This reaction is quantitative and losses stemmed from powder being stuck in filters or under the filter paper. Careful control of activated-carbon quantity was critical: too little left residual iodide that caused a positive iodoform test on compound **(4)** in later steps, disrupting the condensation; too much caused product adsorption, low yield and gray discoloration of product (Entry 2, 35 %). Routine iodoform testing of compound **(4)** before use is strongly recommended for future work.

Table 1: Reaction conditions for hydrolysis from **(2)** to **3**

Experiment No.	n of (2) [mmol]	Equivalents of NaOH	Yield [%]
1	1.00	6	Fail
2	2.52	75	35
3	2.52	75	86
4	6.08	67	96
5	17.7	70	81
6	23	70	84

4.4 First Attempted Synthesis of Salan (6b) via Mannich route

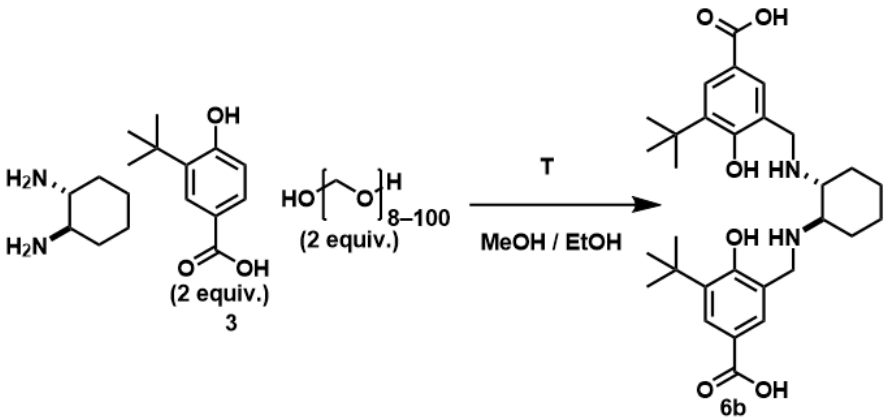


Figure 14: Attempted one-pot Mannich-type double alkylation of **(3)** with paraformaldehyde and *trans*-DACH to give salan ligand **(6b)** directly.

A more direct one-pot Mannich-type double alkylation of **(3)** (2 equiv.) with paraformaldehyde (2 equiv.) and *trans*-DACH (1 equiv.) (Figure 14) with proposed mechanism shown in appendix B.3 was attempted under six conditions (65–100 °C; EtOH or MeOH; Table 2). None yielded uniform solid product. LCMS analysis revealed side products at m/z 332 and 538 (Figure 15), consistent with mono-alkylation followed by intramolecular ring closure between the two amine groups

of DACH rather than double alkylation. **(6b)** is the non-HCl salt version of **(6)** and therefore its problematic solubility might have played a roll which is discussed further down in section 4.7.

Table 2: Reaction conditions for formation of **(6b)**

Experiment No.	T [°C]	Solvent	Time [h]	Additive
1	70	EtOH	4	**
2	65	MeOH	2.5	-
3	70	EtOH	4	*
4	100	H_2O	4	*
5	70	EtOH	18	*
6	100	H_2O	18	*

**: Formaldehyde solution instead of para formaldehyde *: Used methylated version of DACH.

With hindsight, increasing paraformaldehyde to at least 3 equiv. to suppress mono-alkylation, and isolating the product as its HCl salt from the outset to overcome the severe solubility issues of the free amine, may enable this route in future work. This would shorten the overall synthesis by three steps. Though, then there needs to be an additional de-alkylation reaction in order to yield **(6b)** instead of **(8b)** which could face selectivity problems such as getting back just the diamine.

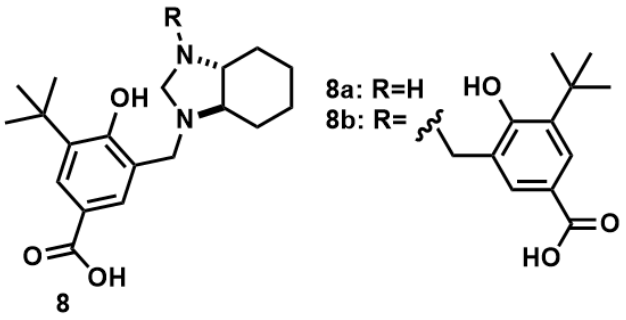


Figure 15: Proposed structures of Mannich side products (**(8a)**, m/z 332; **(8b)**, m/z 538) arising from mono-alkylation and intramolecular ring closure

4.5 Synthesis of 3-tert-butyl-5-formyl-4-hydroxybenzoic acid, (4)

Since the one pot Mannich type reaction was abandoned, a Schiff base condensation reaction was attempted instead; A new formylation step would have to be introduced and also a reduction after the condensation, turning one step into three steps, which is not optimal but it lead to a succesful synthesis of **(7)** in the end albeit not without its difficulties along the way.

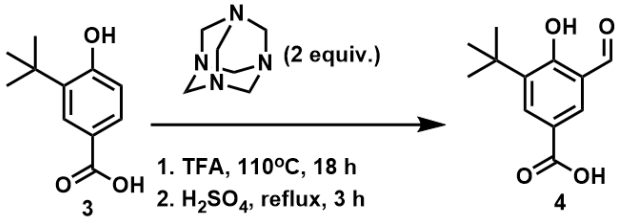


Figure 16: Synthetic scheme to obtain **(4)** from

Duff-type ortho-formylation **(3)** using HMTA (2.0 equiv.) in TFA at 110°C (18 h) followed by H_2SO_4 hydrolysis (reflux, 3 h) gave **(4)** (Figure 16) in 76–82 % yield with different drying conditions (Table 4). First reaction was just a small test and was never dried or quantified, but it was pure according to ^{1}H NMR and thus confirmed selective for the ortho position. The product **(4)** retained large amounts of water owing to its multiple hydrogen-bond-accepting groups, forming a non-Newtonian paste on filtration that required 24 h oven drying each time at 50-70°C. It was found that drying at 70°C

could only be done at most for 1 h, because the Product decomposed and turned pink otherwise. Entry 2 came out fine but entry 3 was left in a 70 °oven for 24 h instead of just for 1 hour and had its appearance changed the next day.

Table 4: Reaction conditions for formation of **4**

Experiment No.	n of (3) [mmol]	Yield [%]	Drying temperature [°C]
1	0.257	-	-
2	2.16	76	70
3	14.4	76	70
4	19.6	82	50

An alternative $MgCl_2$-mediated formylation adapted from[42] (Figure 17) was evaluated but gave only a pink oil. This is likely due to a methyl ester protecting group never being installed to protect the carboxylic acid. This oil began to crystallize after being left in the fumehood for 2 days and when acetone was poured on it, concentric crystals resembling CDs instantly grew which was interesting.

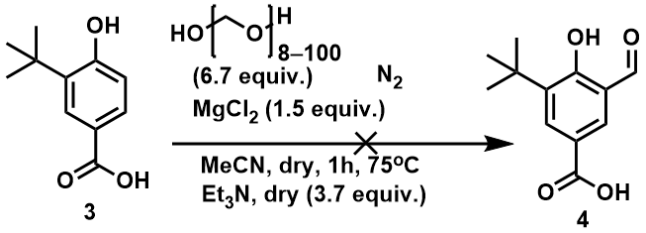


Figure 17: Alternative unsuccessful attempted synthetic scheme for **(4)**: $MgCl_2$-mediated formylation of **(3)**

4.6 Synthesis of Salen type ligand, (5)

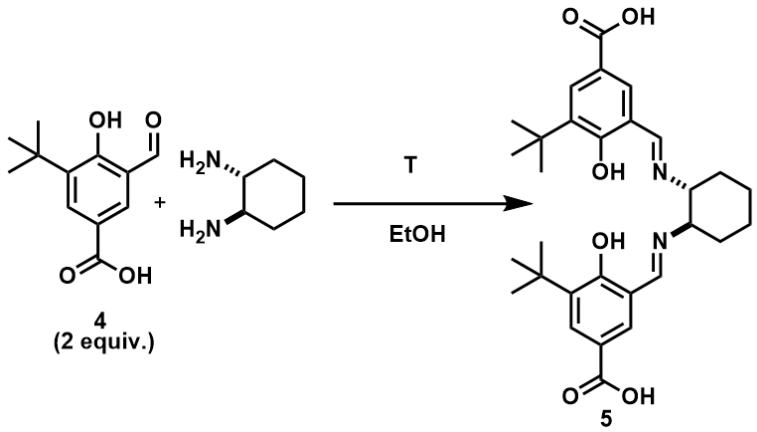


Figure 18: General synthetic scheme to obtain **(5)**: Condensation between **(4)** and *trans*-DACH.

Condensation of **(4)** (2.0 equiv.) with (R,R)-trans-DACH (1.0 equiv.) in ethanol (Figure 18) gave salen 5 in 70 % yield in the single successful experiment (Entry 1 in Table 5: quickly heated up to 70°C, before promptly being cooled overnight while stirring). The imine signal at δ 8.60 ppm in ^{1}H NMR confirmed Schiff base formation. All five subsequent scale-up attempts (Entries 2–6, 80–90 °C, >10 h reflux) yielded tar-like sludge.

Two causes were identified: residual left iodide in **(4)** (a purple color upon addition of EtOH in Entry 5, consistent with a positive iodoform test) promoted side reactions; extended reflux conditions appeared to promote decomposition and/or polymerisation of the ligand. The successful Entry 1 used a notably milder approach: heating to 70 °C for a short initial period, then allowing slow cooling overnight. Column-chromatographic purification of **(4)**4 to remove iodide, and strict replication of Entry 1 conditions at all scales, are the most important recommendations for future attempts. Utilizing column-chromatographic purification for the purification **(4)** would introduce this method to an otherwise column free synthetic route. Another option to keep the synthesis free from columns would be to err on the side of caution and use more rather than less of the activated-carbon during work-up of **(3)** in Section 4.3.

Table 5: Reaction conditions for formation of **5**

Experiment No.	n of (4) [mmol]	T [°C]	Solvent	Yield [%]
1	1.57	70*	EtOH	70
2	0.09	80	EtOH	test
3	0.09	80	glyme	test
4	6.75	80	EtOH	fail
5	3.87	70	EtOH	fail
6	8.00	90	EtOH	fail

4.7 Synthesis of Salan type ligand, (6). Reduction

. Reduction of salen Schiff base **(5)** with $NaBH_4$ (6.5 equiv.) in dry MeOH at 0°C (Figure 19) gave salan HCl **(6)** after

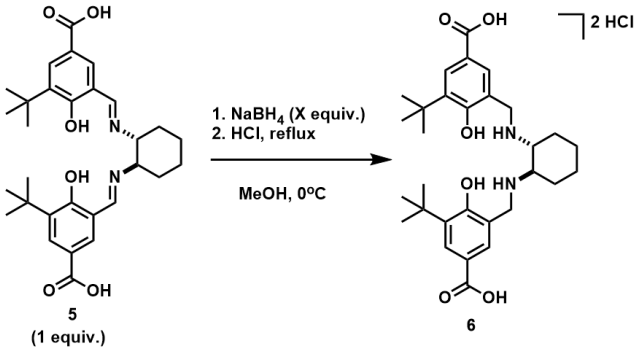


Figure 19: Synthetic scheme to obtain **(6)**

an extensively optimized workup (Table 6). Complete reduction was confirmed by disappearance of the imine signal at δ 8.60 ppm in reaction-control ^{1}H NMR. Borate salt by-products co-precipitated and were difficult to remove. Earlier workup protocols using $NaHCO_3$ for quenching deprotonated the carboxylic acids, causing product loss into the aqueous phase. The successful procedure (Entry 6) involved washing with dilute HCl to form the hydrochloride salt, liquid-liquid extraction with excessive amounts ofEtOAc (**(6)** was insoluble in neutral water and barely soluble in EtOAc), concentration *in vacuo*, refluxing in 3 % HCl (2 h), cooling to below room temperature, filtering, drying in 50°C oven, grinding in mortar with a pestle and finally washing with acetone. Since the HCl salt of the salan is insoluble in acetone while the borate and other impurities dissolve readily, this sequence afforded pure white powder after drying. The overall yield of 35 % reflects accumulated impact of suboptimal starting material purity and the demanding workup. The installation of methyl ester protecting groups on the carboxylic acids in a previous stage is recommended to simplify all downstream steps.

Table 6: Reaction conditions for formation of **6**

Experiment No.	n of (5) [mmol]	Equivalents of $NaBH_4$	Workup & Quenching method
1	0.69	10	$NaHCO_3$, Conc. HCl @ rt
2	0.09	10	$NaHCO_3$, Conc. HCl @ rt
3	0.09	10	$NaHCO_3$, Conc. HCl @ rt
4	0.10	6.5	10% HCl, H_2O @ rt, Et_2O & EtOAc
5	0.19	6.5	Conc. HCl, water @ rt, EtOAc
6	3.20	6.5	-II-, 3 M HCl Reflux, acetone wash

4.8 Synthesis of CoSalan complex, (7)

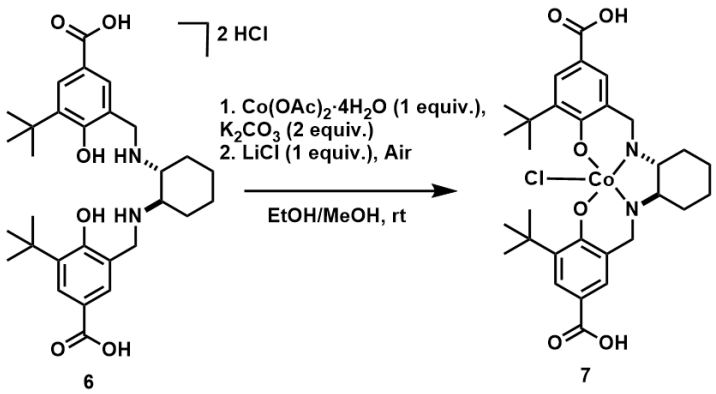


Figure 20: Synthetic scheme to obtain **(7)**

Cobalt complexation was investigated using two cobalt sources under five different sets of conditions (Table 7). The target oxidation state was Co(III), required for subsequent photo-induced reduction to Co(II) and hydride formation as seen in figure 6. Entries 1 and 2 used $Co(OAc)_2$ at elevated temperature (80–85 °C) based upon a previous reported method[43] gave dark, poorly precipitated products with evidence of cobalt nanoparticle formation. Entry 3, based upon a different reported method[44] ($CoCl_2$ with H_2O_2 as oxidant, 150°C, Figure 21) caused near-complete decomposition (ca. 1 % mass yield); an interesting solvatochromic colour change from mauve to bright blue at 150 °C, reversing on cooling, was observed. For each of the entries that used $Co(OAc)_2$, the pink salt initially turned bright turquoise when the first drops made contact with the solution. and then subsequently turned dark, either brown or green. Each of these were green when the solvent was removed. Entry 4 and 5, following the room-temperature protocol of Zhang et al.[24] for an analogous CuSalan complex, each gave more of the dark green precipitate, without as much cobalt nanoparticle dust and with entry 5 giving the most. Dark green is the expected and desired color because of how cobalt changes the electron density and delocalization. Entry 5 was assessed as the best result and added K_2CO_3 (2 equiv.) to deprotonate the phenols fully before cobalt addition, which is mechanistically appropriate and resulted in visibly more precipitate.

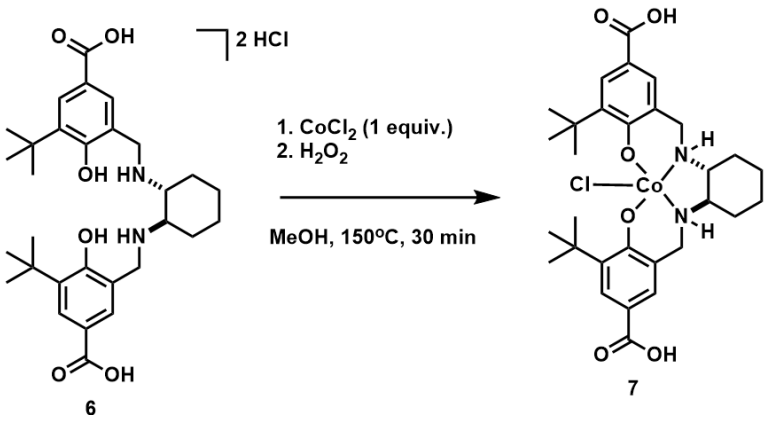


Figure 21: Alternative attempted synthetic scheme for **(7)**

Table 7: Reaction conditions for formation of **(7)**. Each was done at 0.08 mmol scale.

Entry	Type of Co-salt	time [hours] and T [°C]	Solvent	Extras
1	$Co(OAc)_2$ (2 eq.)	4h @ 85°C + 4h @ 50°C	MeOH	LiCl (6 eq.)
2	$Co(OAc)_2$ (1 eq.)	4h @ 80°C + 4h @ 50°C	EtOH & MeOH	LiCl (6 eq.)
3	$CoCl_2$ (1 eq.)	0.5h @ 150°C + 5h @ rt	MeOH	H_2O_2 (30%, 0.3 ml)
4	$Co(OAc)_2$ (1 eq.)	96h @ rt	EtOH & MeOH	LiCl (1 eq.)
5	$Co(OAc)_2$ (1 eq.)	96h @ rt	EtOH & MeOH	LiCl, K_2CO_3 (2 eq.)

FTIR analysis of Entries 3, 4, and 5 in comparison with the free ligand HCl salt **(6)** (Figure 22) shows a clear shift of the N-H stretching vibration from 3220 cm^{-1} (secondary amine in **(6)**) to 3410 cm^{-1} in Entries 4 and 5, consistent with reduced

electron density on nitrogen upon cobalt coordination and constituting preliminary evidence for successful complexation. Entry 3 does not show this shift, consistent with decomposition. Entry 5 was recrystallized in MeOH which required 50 ml, a substantial amount. It was still not clear so it was hot filtered (around 30-50°C) which left some black residue on the filter. Then the solution was clear still dark green. It was then concentrated by heating to 50°C in a water bath while simultaneously blowing on it with N_2 gas until it °started to crash out. It was then sonicated then left in -21°C freezer for a week, which yielded some crystals. Formal confirmation of the Co(III) oxidation state requires EPR spectroscopy, cyclic voltammetry, and/or high-resolution MS, none of which were available within the project period. Crystal growth by Et_2O/DMF vapor diffusion is recommended for future SCXRD structure determination.

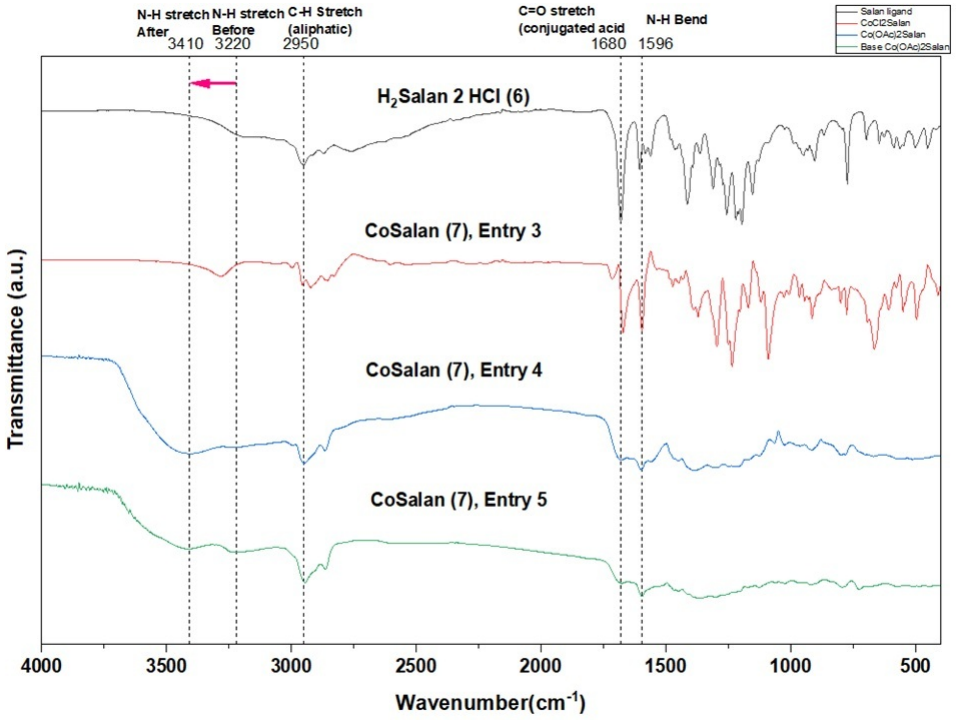


Figure 22: Comparison of peaks between FTIR spectra of **(6)** and three entries of **(7)**. Top two spectra were performed neat, and bottom two spectra were performed with KBr pellets.

5 Conclusions

A multi-step column-free synthesis of the CoSalan complex **(7)**, rationally designed as a bi-functional racemic chiral linker for MOF-based photocatalytic ATH, was successfully developed. The method is applicable to yield the enantiopure linker, provided either of the *trans*-DACH enantiomers were used as starting materials during the condensation reaction to form compound **(5)**. All synthetic intermediates (compounds **(1–6)**) were characterized by NMR, with the salan HCl salt **(6)** and **(7)** both being characterized by FTIR. Preliminary FTIR data support successful cobalt complexation in Entries 4 and 5 of the complexation step. MOF synthesis was not achieved within the project period.

The main hurdles encountered during synthesis of **(7)** were: the persistent iodide contamination originating from the hydrolysis of **(2)**, propagating into and disrupting the Schiff base condensation; sensitivity of the condensation (from **(4)** to **(5)**) to temperature and scale, with only the mildest conditions proving successful; demanding purification of salan hydrochloride **(6)**, requiring a specific HCl-reflux/acetone-wash sequence; and incomplete characterization of the cobalt complex 7, in particular the absence of oxidation-state-specific analytical data.

5.1 Future Work

The following modifications are recommended for continuation of this work:

Iodide removal (compound (3)). Implement routine iodoform testing on compound **(4)** before use in the condensation. If positive, re-purify by repeated activated-carbon treatment or ion-exchange chromatography.

Protecting group strategy. Introduce methyl ester groups on the carboxylic acids from the outset to simplify all downstream steps and permit standard workup protocols. Deprotect at a late stage before MOF synthesis.

Condensation (compound (5)). Use 70°C in the beginning with slow cooling (replicating Entry 1).

Cobalt complex (7) characterization. Confirm the Co(III) oxidation state by EPR spectroscopy. Determine the molecular formula by high-resolution MS. Measure the redox potential by cyclic voltammetry. UV-vis spectroscopy with linear regression to monitor complex formation. Grow single crystals by Et_2O/DMF vapour diffusion.

MOF synthesis. Once pure, fully characterized linker **(7)** is available, conduct solvothermal screening with Zr, Al, and Bi metal salts under various conditions. Characterize resulting materials by PXRD, TGA, SEM and gas sorption analysis.

Photocatalytic evaluation. Mechanistic studies using TEMPO as a radical trap. to evalueate ground-state reactivity compared to visible light excitation. Evaluation of photocatalytic versus ground-state reactivity to establish whether visible light excitation is essential.

ATH scope and enantioselectivity studies. This will be done on substrates including pro-chiral ketones and sulfonamides. Enantiomeric excess should be evaluated along with recyclability and stability of catalyst. Chiral resolution of *trans*-DACH by crystallization together with chiral additive to access the enantiopure MOF (starting with the racemate to save resources). Recyclability and scalability assessment.

6 Acknowledgements

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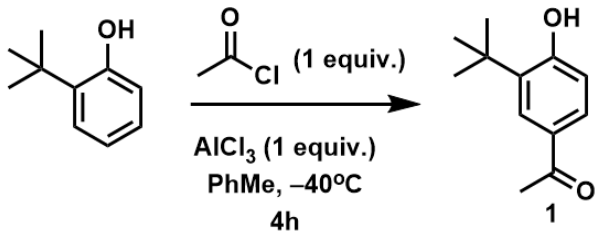
7 Experimental section

7.1 Methods and materials

All chemicals and reagents were purchased commercially from *Chemtronica* and used without further purification. NMR spectra were acquired at 25°C and were performed on an *Agilent MR-400-DD2* spectrometer (400 MHz for ^{1}H), 101 MHz for ^{13}C1) equipped with OneNMR probes. The chemical shifts of the spectra were reported in ppm (δ scale) and referenced using the residual solvent peaks of either DMSO-d6(δ_H = 2.50 ppm, δ_C = 39.52 ppm) or $CDCl_3$(δ_H = 7.26 ppm). FTIR spectra were measured on a Bruker Tensor 27 spectrometer using Opus 7.0 software. Samples were either measured neat or prepared in KBr discs.

7.2 Synthesis

Compound (1): 1-(3-(*tert*-butyl)-4-hydroxyphenyl)ethan-1-one



Compound **1** was synthesized according to an adapted literature procedure[45]. $AlCl_3$ (6.70 g, 50.0 mmol, 1 equiv.) was added to 200 ml toluene under nitrogen stirring at -40°C before adding 2-*tert*-butyl-phenol (7.68 ml, 50.0 mmol, 1 equiv.) drop-wise and left stirring for 1.5 h. AcCl (3.55 ml, 50.0 mmol, 1 equiv.) was then added drop wise and left stirring for 2.5 h. Reaction mixture was then poured into ice / water mixture and left stirring for 1 hour. Precipitate was filtered and collected, then dried overnight in 50°C oven. Then 4.07 g product was collected at 42% yield.

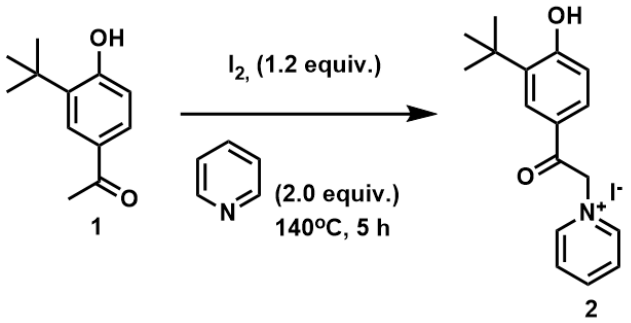
^{1}**H NMR** (400 MHz, $CDCl_3$) δ 7.95 (d, *J* = 2.2 Hz, 1H, Ar-H), 7.72 (dd, *J* = 8.3, 2.2 Hz, 1H, Ar-H), 6.71 (d, *J* = 8.2 Hz, 1H, Ar-H), 5.32 (s, 1H, ArO-H), 2.55 (s, 3H, CH_3), 1.43 (s, 9H, *t*Bu-H).

^{1}**H NMR** (400 MHz, DMSO-d6) δ 10.29 (s, 1H, ArO-H), 7.74 (d, J = 2.2 Hz, 1H, Ar-H), 7.66 (dd, J = 8.3, 2.2 Hz, 1H, Ar-H), 6.82 (d, J = 8.3 Hz, 1H, Ar-H), 2.43 (s, 3H, CH_3), 1.32 (s, 9H, *t*Bu-H).

^{13}**C NMR** (101 MHz, DMSO-d6) δ 135.21, 128.57, 126.83, 115.83, 40.15, 39.94, 39.73, 39.52, 39.31, 39.10, 38.90, 34.32, 29.08, 26.20

Analytical data matched literature values[39].

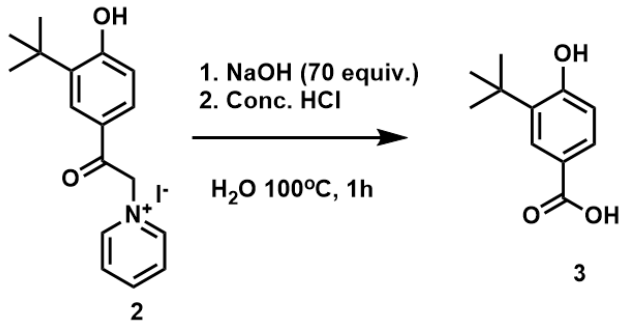
Compound (2): 1-(3-(tert-butyl)-4-hydroxyphenacyl)pyridinium iodide



To a solution of I_2 (6.45 g, 25.4 mmol, 1.2 equiv.) in 42.3 ml of pyridine (2 ml/1 mmol of **1**, 2 equiv.), **1** (4.07g, 21.2 mmol, 1.0 equiv.) was added and heated to 140°C under stirring for 5 hours. It was allowed to cool down to room temperature before concentrating *in vacuo*. Resulting crude was triturated with Et_2O then recrystallized from wet and left in fridge over weekend. Recrystallized product was filtered, washed with water to yield 7.071 g golden-orange crystals at a yield of 84%.

^{1}H NMR (400 MHz, DMSO-d6) δ 10.76 (s, 1H, ArO-H), 8.95 (d, *J* = 6.4 Hz, 2H, Py-H), 8.72 (t, *J* = 7.8 Hz, 1H, Py-H), 8.25 (dd, *J* = 7.9, 6.3 Hz, 2H, Py-H), 7.84 (d, *J* = 2.2 Hz, 1H, Ar-H), 7.82 (dd, *J* = 8.3, 2.2 Hz, 1H, Ar-H), 7.00 (d, *J* = 8.3 Hz, 1H, Ar-H), 6.39 (s, 2H, CH_2), 1.39 (s, 9H, *t*Bu-H).

Compound (3): 3-tert-butyl-4-hydroxybenzoic acid

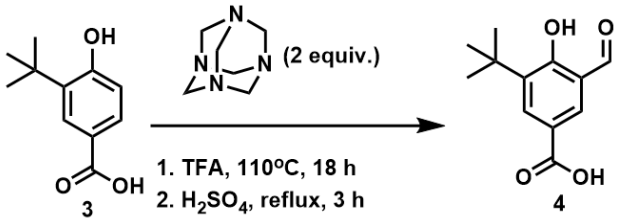


In a 1 l round bottom flask to a solution of **2** (9.1 g, 23 mmol, 1 eq) in 710 ml water stirring at 100°C NaOH (84 g, 2100 mmol, 90 eq.) was added which made the solution turn bright orange color at first then black. It was left to reflux for 1 hour before cooling down to ambient temperature. Then slightly more than 2 spatulas worth of activated carbon was added. Solution was filtered through cotton before it was acidified with Conc. HCl (200 ml, 37%). Then it was left to stir for 1 hour. Precipitate was filtered then dried in 50°C oven overnight. Then 3.8 g product was collected at 84% yield.

^{1}H NMR (400 MHz, DMSO-d6) δ 12.34 (s, 1H, COO-H), 10.20 (s, 1H, ArO-H), 7.79 (d, *J* = 2.1 Hz, 1H, Ar-H), 7.64 (dd, *J* = 8.4, 2.1 Hz, 1H, Ar-H), 6.84 (d, *J* = 8.4 Hz, 1H, Ar-H), 1.35 (s, 9H, *t*Bu-H).
Analytical data matched literature values[10].

^{13}C NMR (101 MHz, DMSO-d6) δ 167.51, 160.25, 135.04, 128.95, 128.14, 115.91, 40.15, 39.94, 39.73, 39.52, 39.31, 39.10, 38.89, 34.25, 29.10.

Compound (4): 3-tert-butyl-5-formyl-4-hydroxybenzoic acid



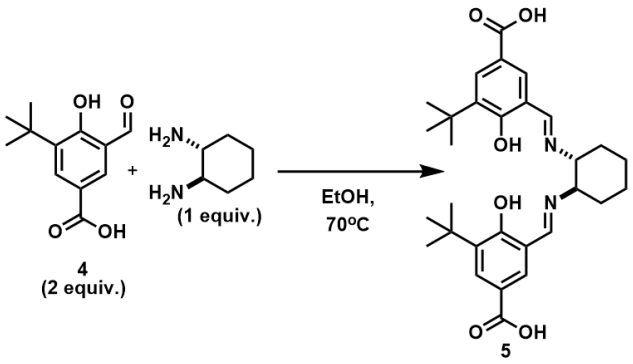
Compound **4** was synthesized according to an adapted literature procedure[10]. Compound **3** (3.80 g, 19.6 mmol, 1.0 equiv.) and HMTA (5.49 g, 39,1 mmol, 2.0 eq.) were charged into 500 ml round bottom flask before adding 154 ml TFA. Solution stirred at 85°C while refluxing for 18 hours. H_2SO_4 (33%(aq), 123 ml) was then added stirring for an additional 3 hours at reflux before cooling down to room temperature. Reaction solution was extracted twice with Et_2O. Combined organic phase was washed with water and brine and dried over $MgSO_4$, filtered over to round bottom flask, concentrated under reduced pressure. Crude had crystallized and was orange. Crude was drop wise dissolved in 2M NaOH. Solid impurities were filtered off over vacuum before filtrate was poured into E-flask containing 10% HCl and ice mixture forming precipitate with curdled appearance. Precipitate was filtered, washed with water/ HCl and then put in 50°C oven overnight. Then 3.56 g product was collected at 82% yield.

^{1}H NMR (400 MHz, DMSO-d6) δ 12.95 (s, 1H, COO-H), 12.25 (s, 1H, ArO-H), 10.07 (s, 1H, Ar-CHO), 8.30 (d, *J* = 2.1 Hz, 1H, Ar-H), 8.08 (d, *J* = 2.1 Hz, 1H, Ar-H), 1.40 (s, 9H, *t*Bu-H).

Analytical data matched literature values[10].

Compound (5): Salen ligand

In a solution of **4** (350 mg, 1.57 mmol, 2.0 equiv.) in 20 ml EtOH stirring at 70°C with reflux, *trans*-DACH (130 mg, 0.787 mmol, 1.0 equiv.) dissolved in 10 ml EtOH was added drop wise. Reaction was allowed to cool down to ambient temperature and left to stir over weekend. Reaction mixture was concentrated in vacuo. Solid was scraped of with Et_2O

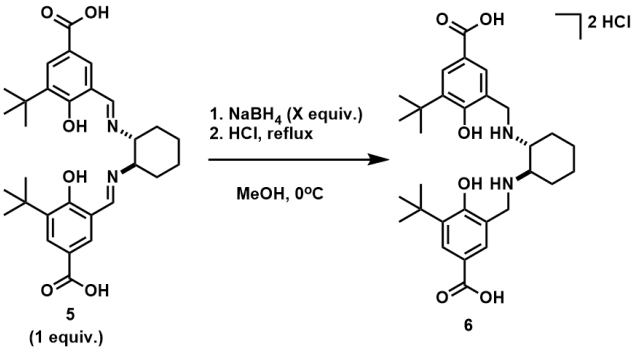


and collected into a falchion tube which was centrifuged at 3400 rpm for 10 min twice, removing and adding more Et_2O in between. Falchion tube was put in 70°C oven to dry. 358 mg product was collected, which was 80% pure from NMR at 70% yield.

^{1}**H NMR** (400 MHz, DMSO-d6) δ 8.60 (s, 2H, -CH=N-), 7.80 (d, J = 2.1 Hz, 2H, Ar-H), 7.76 (d, J = 2.2 Hz, 2H, Ar-H), 3.60 (s, 2H, cHex-H), 2.01 – 1.47 (m, 8H, cHex-H), 1.31 (s, 18H, *t*Bu-H).

Analytical data matched literature values[10].

Compound (6): Salan dihydrochloride

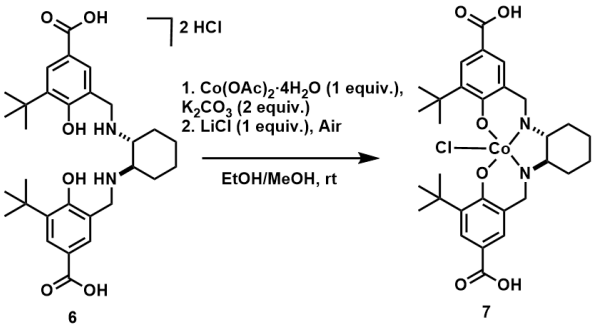


Compound **(6)** was synthesized according to a modified literature procedure[24]. Compound **5** (1.67g, 3.2 mmol, 1 equiv.) was dissolved in 30 ml dry MeOH at 0°C in an ice bath under stirring. Then $NaBH_4$ (0.8 g, 21 mmol, 6.5 equiv.) was added portionwise. Reaction turned from yellow to clear light yellow. after 2 hours some more $NaBH_4$ was added and reaction control proton NMR was performed showing absence of imine peak at δ 8.60 ppm. Solution was then concentrated in vacuo, then dissolved in EtOAc. It was then made into HCl salt by washing EtOAc with HCl and water. Then aqueous phase was extracted more with EtOAc. Combined organic phase was dried over Na_2SO_4 then because product did not fully dissolve in organic phase it was decanted over to round bottom flask and concentrated *in vacuo* portion wise. Resulting crude was refluxed in 20 ml 3% HCl for 2 hours then left to cool down in fridge. Sludgy brown caramell-y solid was filtered and washed with cold water, then put in 50°C oven to dry overnight. Crude was collected as light brown brittle glass-like shards. Crude was ground to off-white powder using mortar and pestle before being washed with acetone and filtered of. Powder was put in 50°oven overnight. 680 mg white powder was collected at a yield of 35%.

^{1}**H NMR** (400 MHz, DMSO-d6) δ 12.68 (s, 2H, COO-H), 9.68 (s, 2H, ArO-H), 9.29 (s, 2H, NH_2), 8.86 (s, 2H, NH_2), 8.01 (s, 2H, Ar-H), 7.86 (s, 2H, Ar-H), 4.36 (d, *J* = 76.3 Hz, 4H, N-CH_2), 1.81 (s, 4H, cHex-H), 1.39 (s, 4H, cHex-H), 1.37 (s, 18H, *t*Bu-H), 1.35 (s, 2H, cHex-H).

Compound (7): CoSalan complex

Compound **6** (50 mg, 0.08 mmol, 1 equiv.) was dissolved in 0.5 ml EtOH in a microwave vial. K_2CO_3 (23 mg, 0.17 mmol, 2 equiv.) was added to it while stirring at room temperature for an hour. The vial was purged and saturated with N_2 gas. K_2CO_3 did not fully dissolve. Solution had turned from clear light yellow to cloudy light yellow. Then $Co(OAc)_2 \cdot 4H_2O$ (21 mg, 0.08 mmol, 1 equiv.) dissolved in 2 ml MeOH was added dropwise which turned from bright pink to bright turquoise at first, then deep blue then back to pinkish brown with clear precipitate. Then it was cloudy thick greyish brown and was left to stir overnight. Then LiCl (3.5 mg, 0.08 mmol, 1 eq.) dissolved in 0.3 ml MeOH was added to



reaction. Then the microwave vial was purged with air by bubbling it and left to stir for 1 hour before being left in fridge over weekend. Green precipitate was gravity filtered and collected as a wet crude of 58 mg, yield of 112%. Crude was redissolved in 50 ml MeOH and sonicated before hot filtrating away black solids yielding a clear green solution, which was concentrated by heating to 50°C in a water bath while blowing N_2 gas until it started to crash out yielding a super concentrated solution which was put in -21°C Freezer for 7 days. Full characterization requires EPR, high-resolution MS, cyclic voltammetry, and SCXRD.

A Appendix

A.1 NMR Spectra

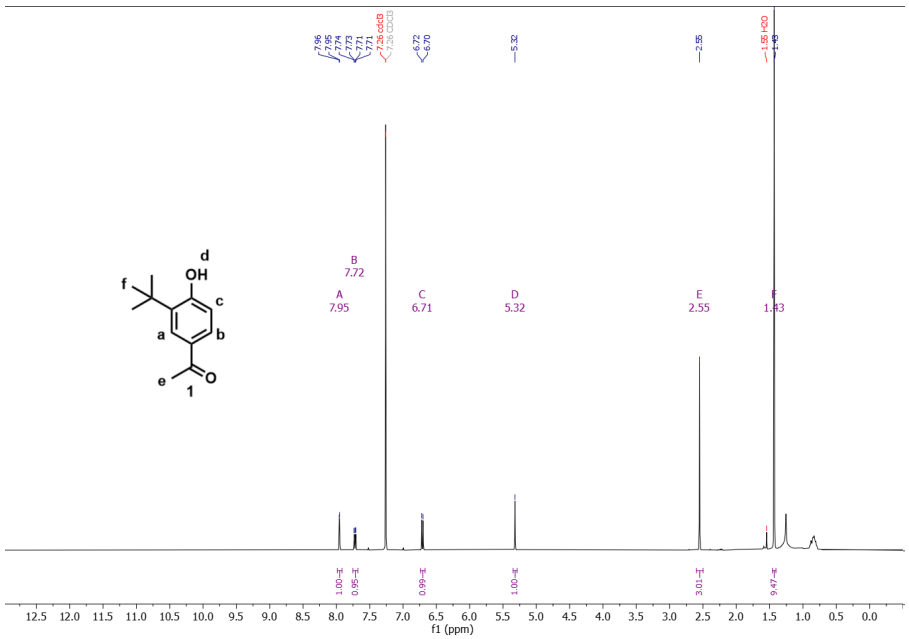


Figure A.1: ^{1}H NMR (400 MHz, $CDCl_3$) spectrum of **(1)**.

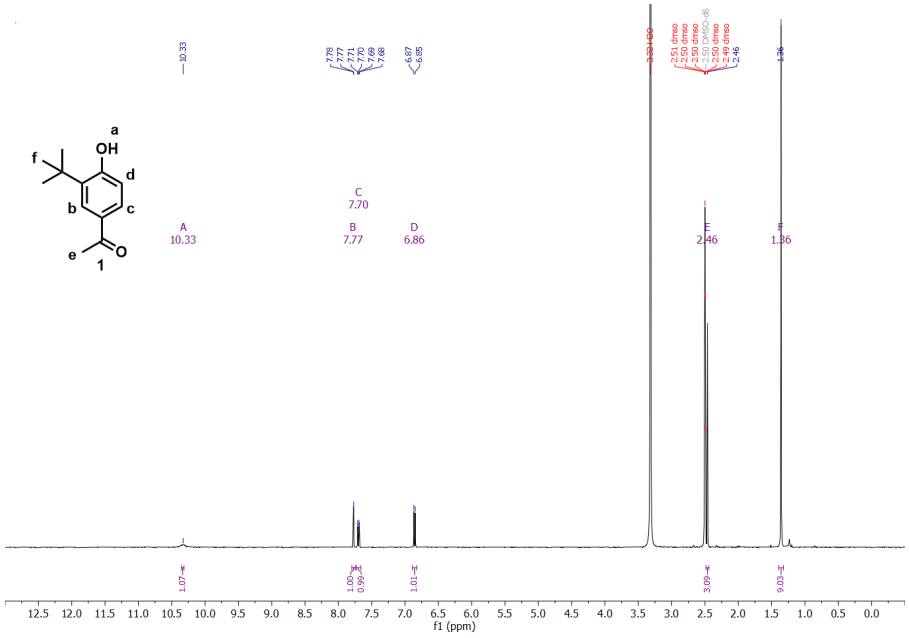


Figure A.2: ^{1}H NMR (400 MHz, DMSO-d6) spectrum of **(1)**.

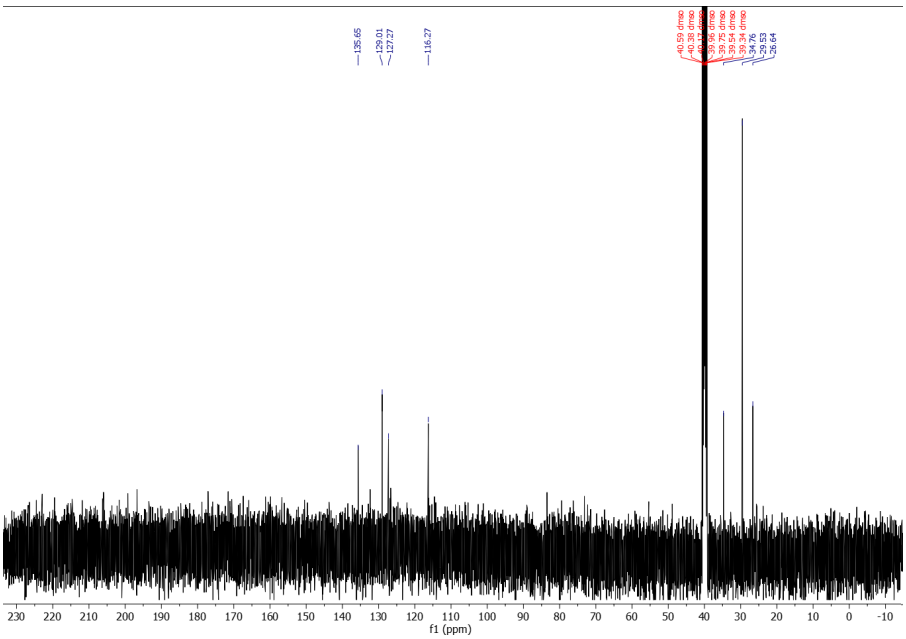


Figure A.3: ^{1}3C NMR (101 MHz, DMSO-d6) spectrum of **(1)**.

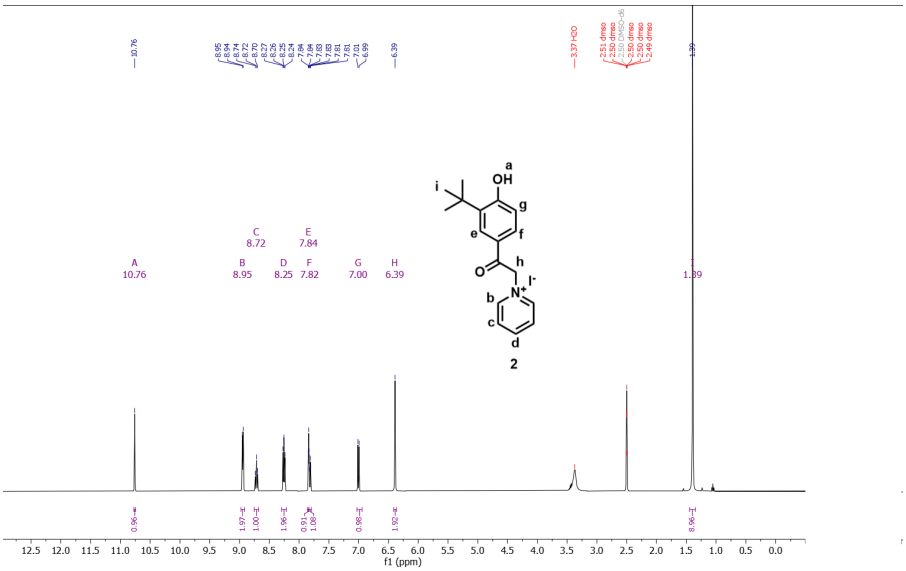


Figure A.4: ^{1}H NMR (400 MHz, DMSO-d6) spectrum of **(2)**.

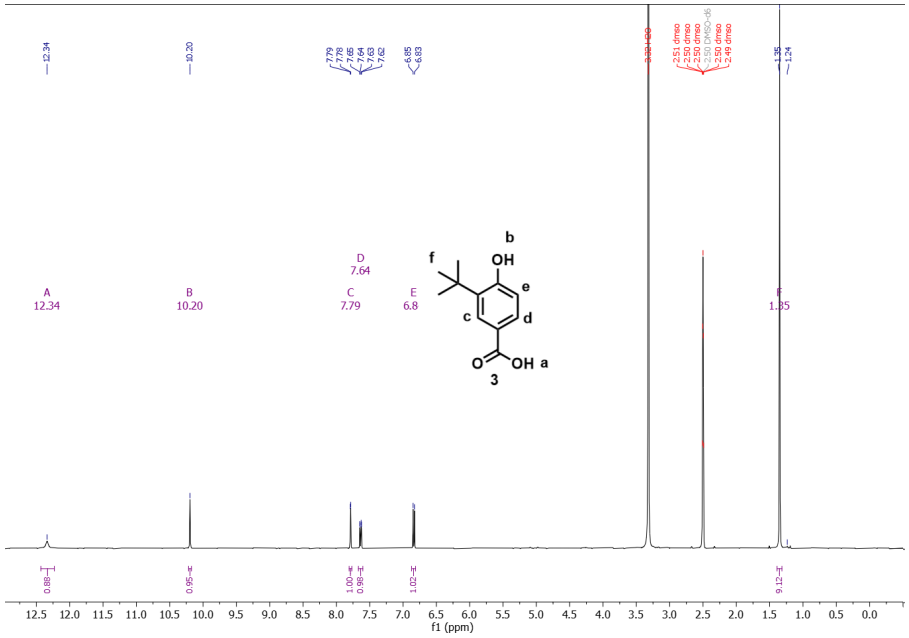


Figure A.5: ^{1}H NMR (400 MHz, DMSO-d6) spectrum of **(3)**.

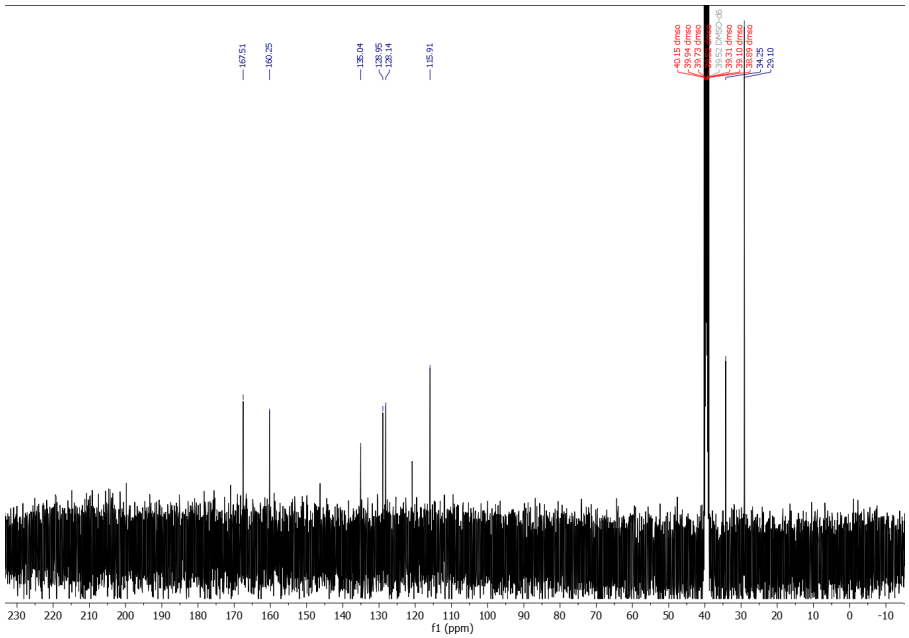


Figure A.6: ^{1}3C NMR (101 MHz, DMSO-d6) spectrum of **(3)**.

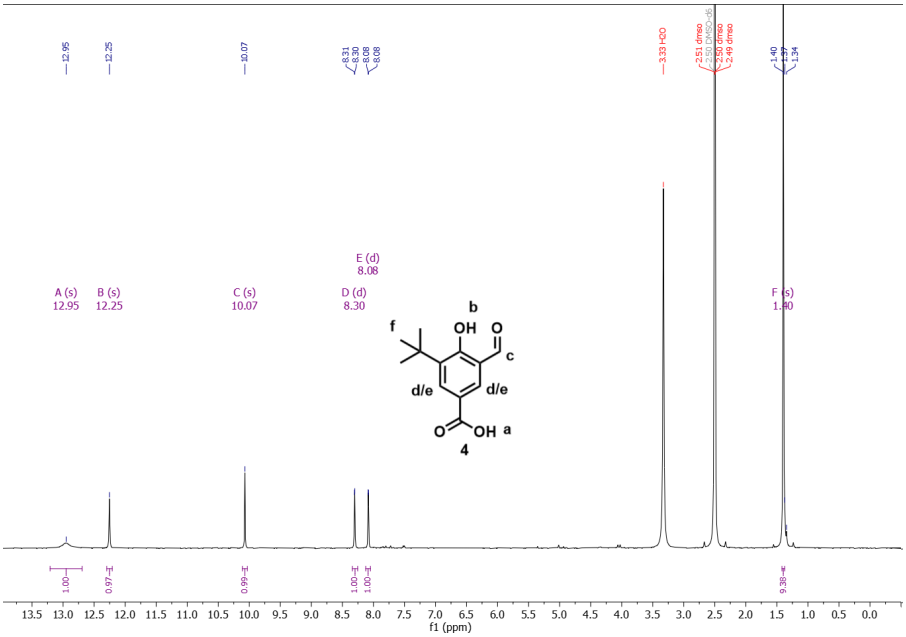


Figure A.7: ^{1}H NMR (400 MHz, DMSO-d6) spectrum of **(4)**.

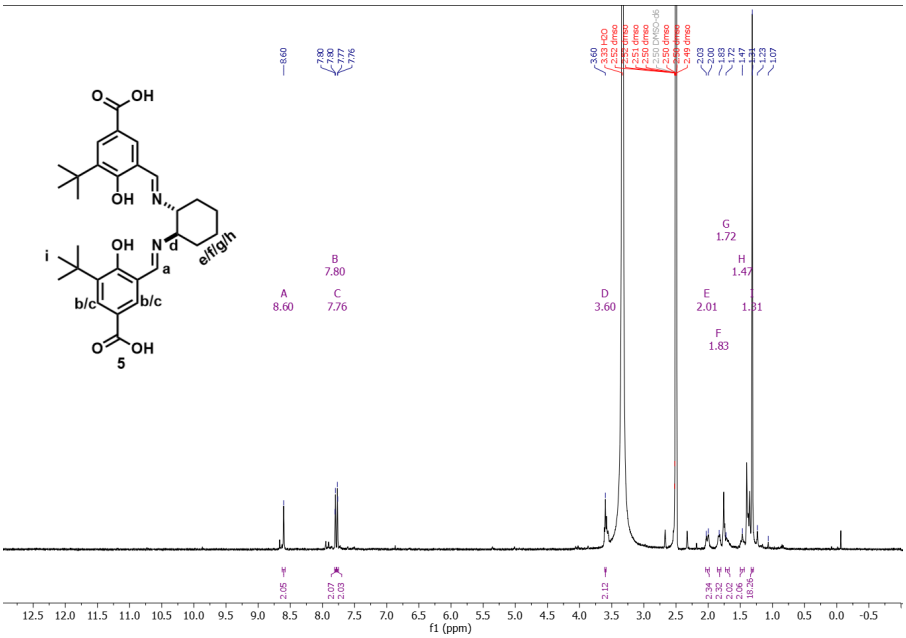


Figure A.8: ^{1}H NMR (400 MHz, DMSO-d6) spectrum of **(5)**.

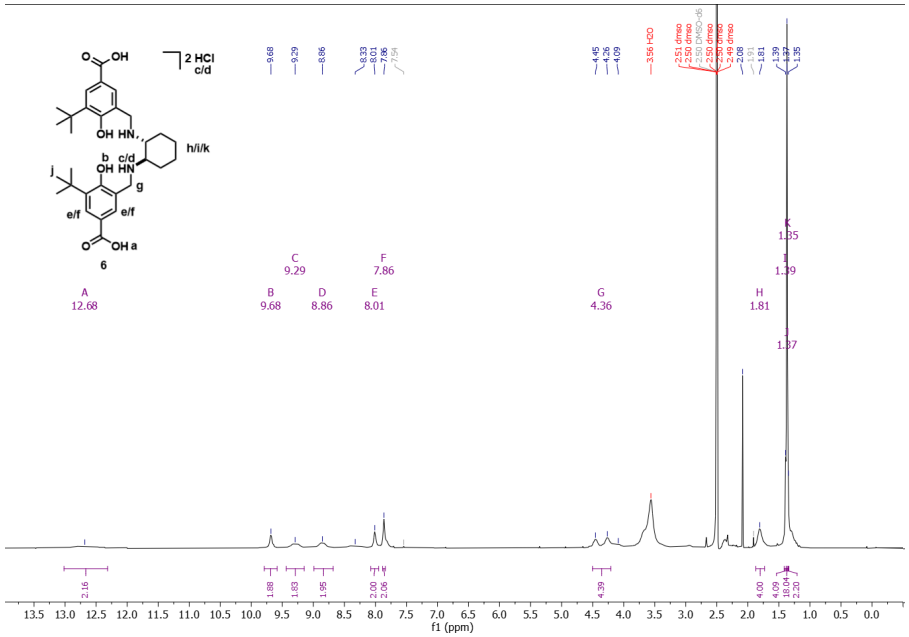


Figure A.9: ^{1}H NMR (400 MHz, DMSO-d6) spectrum of **(6)**.

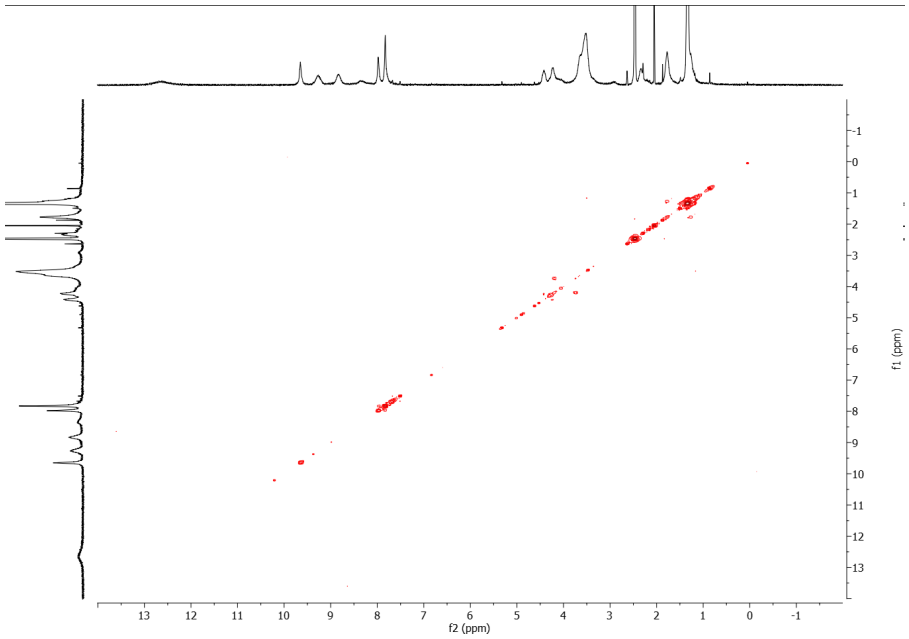


Figure A.10: gCOSY NMR (400 MHz, DMSO-d6) spectrum of **(6)**.

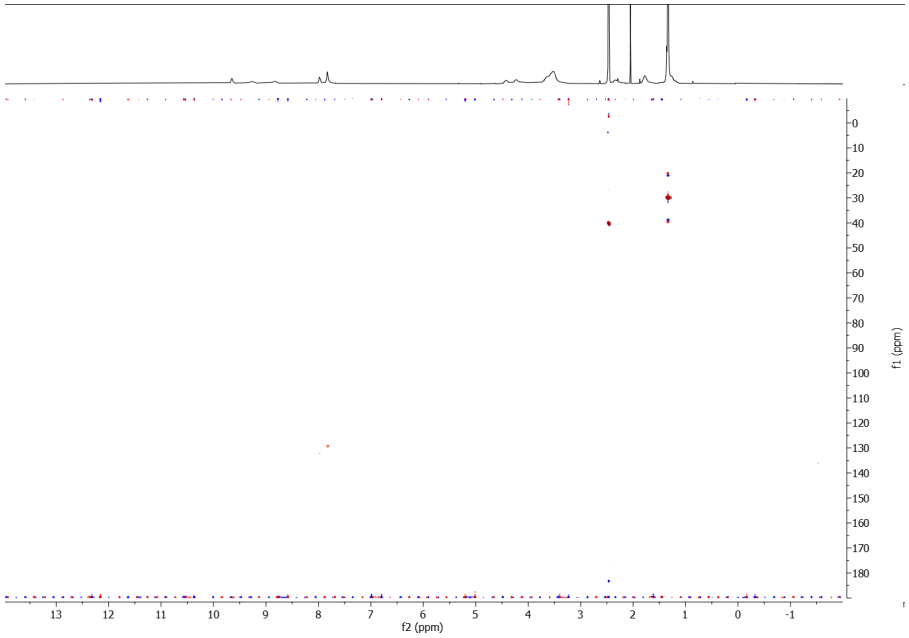


Figure A.11: gHSQC NMR (400 MHz & 101 MHz, DMSO-d6) spectrum of **(6)**.

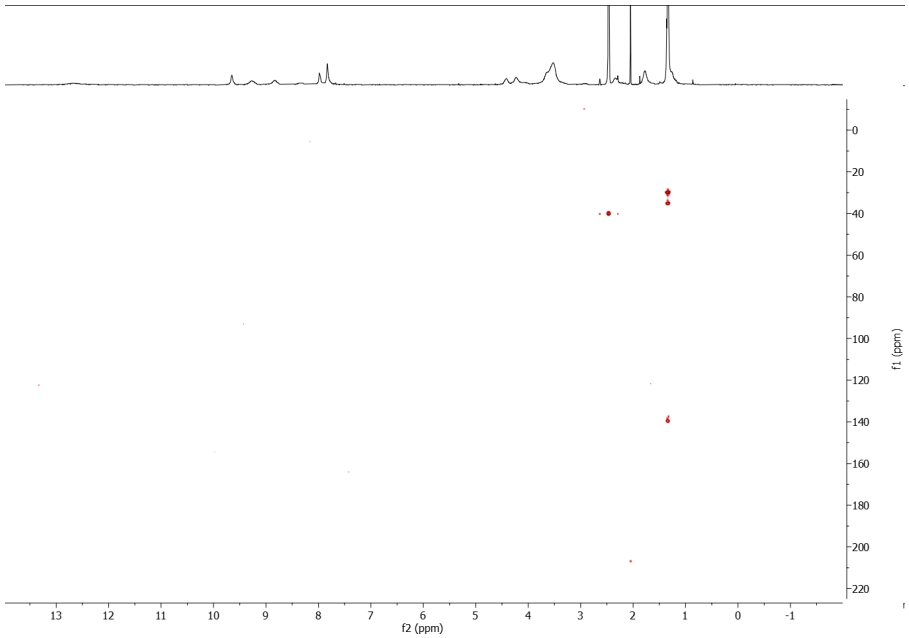


Figure A.12: gHMBC NMR (400 MHz & 101 MHz, DMSO-d6) spectrum of **(6)**.

A.2 FTIR Spectra

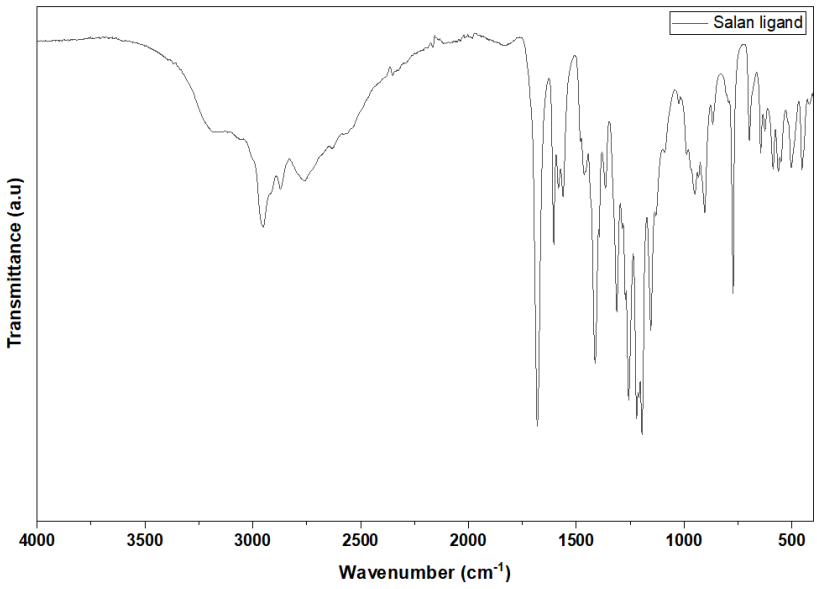


Figure A.13: FTIR spectrum of **(6)**

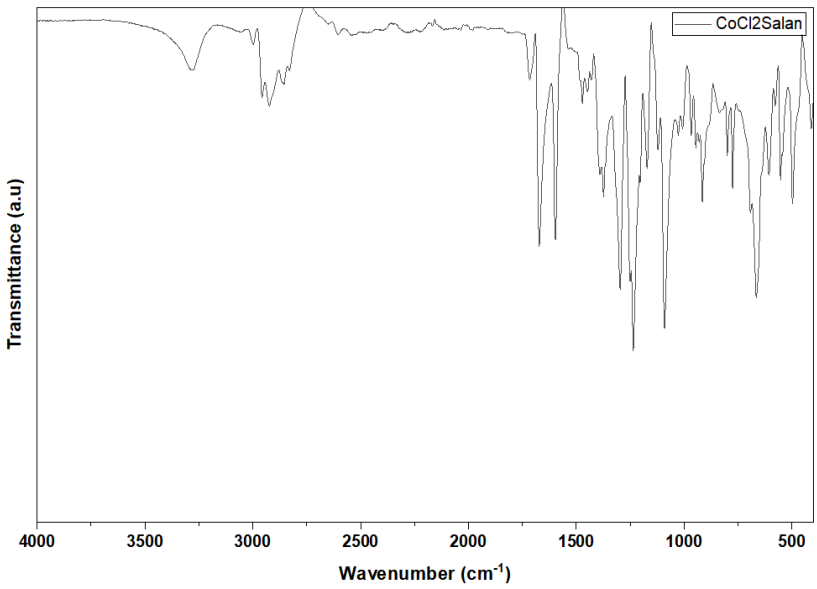


Figure A.14: FTIR spectrum of **(6)**, $CoCl_2$ pathway

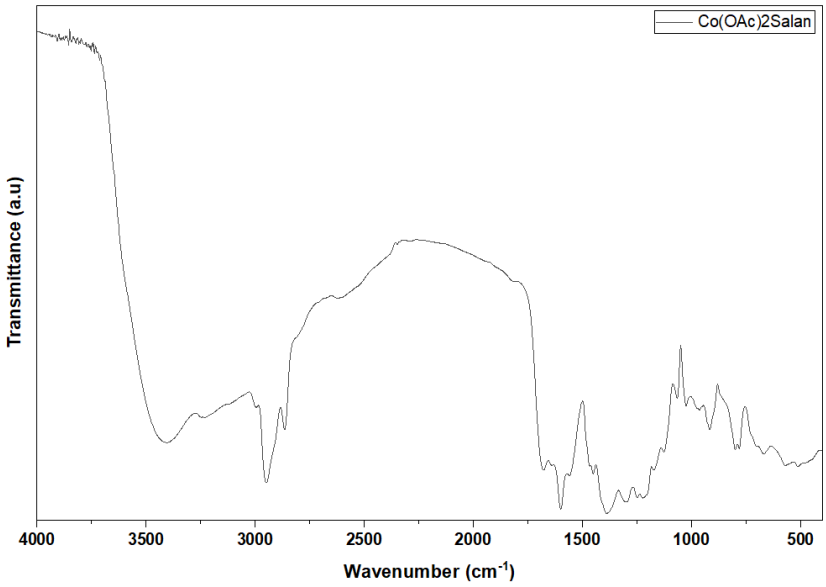


Figure A.15: FTIR spectrum of **(6)**, $Co(Oac)_2$ pathway at room temperature

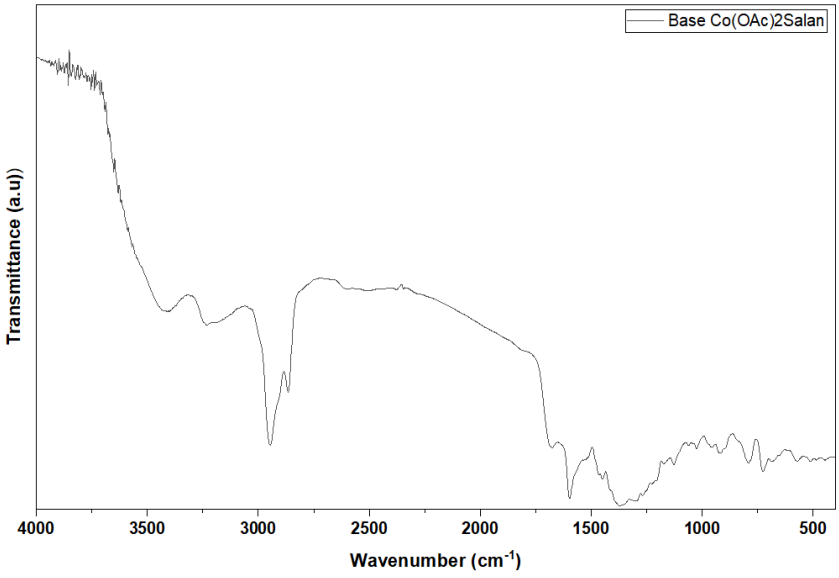


Figure A.16: FTIR spectrum of **(6)**, $Co(Oac)_2$ pathway at room temperature with 2 equivalents base

B Appendix

B.1 Mechanism of formation of pyridinium iodide salt (2)

The mechanism for formation of pyridinium iodide salt **(2)** from compound **(1)]**, I_2, and pyridine at 140°C is shown in Figure B.1, adapted from Pearson et al.[41] and modified according to observed results. The reaction is analogous to the iodoform reaction, with pyridine acting as both base and nucleophile. Using 2 equivalents of pyridine as the only solvent without water gives the desired pyridinium salt. If excess pyridine and water are both present, the carboxylic acid **(3)** can form directly in the first step, but in an uncontrolled manner.

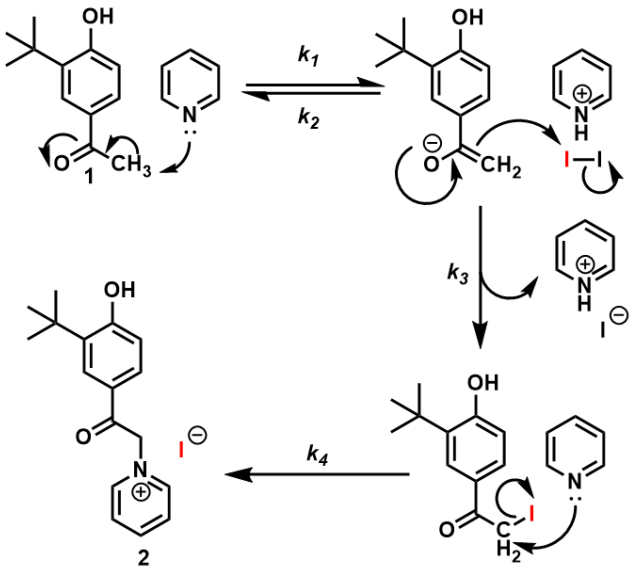


Figure B.1: Reaction mechanism of formation of pyridinium iodide salt (**2**)

B.2 Mechanism of basic hydrolysis to carboxylic acid (3)

To make the carboxylic acid (**3**) in a controlled matter excess NaOH was added in a separate second step instead of the pyridine acting as the base in the previous reaction step and thus the reaction mechanism was changed accordingly (Figure B.2) from the mechanism proposed by Pearson et al.[41], which required 2 equivalents of I_2. The large excess of base (67–90 equiv.) was required to drive the reaction to completion and ensure that the carboxylate remained fully deprotonated throughout, preventing re-formation of side products.

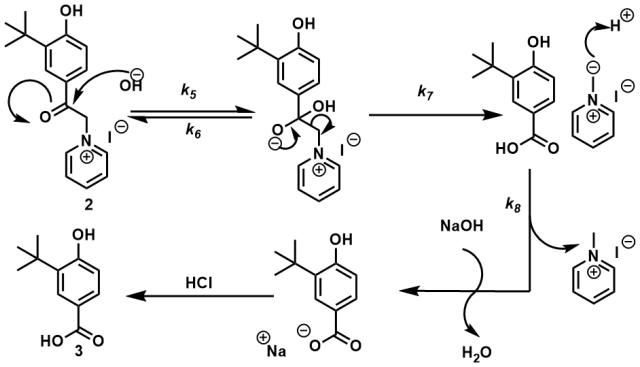


Figure B.2: Reaction mechanism of basic hydrolysis of pyridinium iodide salt to get carboxyllic acid **(3)**

B.3 Mechanism of formation of salan (6b), alternative failed synthesis

The proposed double Mannich-type one pot condensation reaction would hav saved 2 whole steps and was postulated to follow the mechanism shown in figure B.3. It did not go as planned because formation of side products, with their structures seen in figure 15.

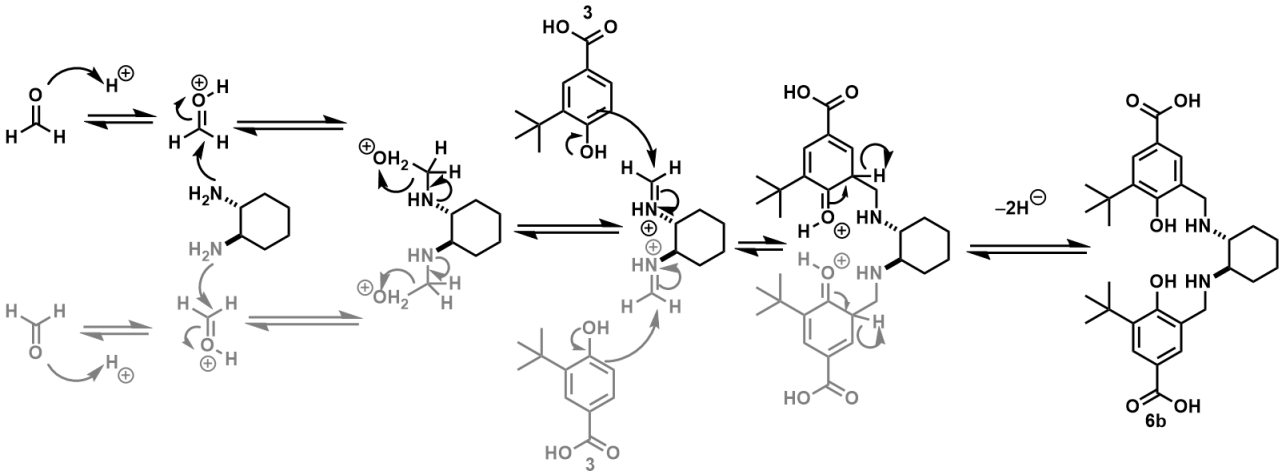


Figure B.3: Double Mannich reaction, failed in practice due to formation of side products